

PDE HOMEWORK 2024

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文档说明 本文档为 PDE 作业的参考答案

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一. PDE HOMEWORK 1

1.1 章一第 9 题

考虑三维长杆, 其中 XOY 为杆的截面, 且 Z 为热传导方向。

$$\Delta Q = C_m \Delta T = \int_D C \rho (u|_{t_2} - u|_{t_1}) dx dy dz = \int_D \int_{t_1}^{t_2} C \rho \frac{\partial u}{\partial t} dt dx dy dz, \quad (1)$$

其中 $D = \{(x, y, z) : x^2 + y^2 \leq R^2, z \geq 0\}$, ρ 是密度, C 是比热容。

由边界进入的热量为

$$Q_{in} = \int_{t_1}^{t_2} \int_{\partial D} \nabla \cdot \vec{n} dS dt = \int_{t_1}^{t_2} \int_{\partial D} k \frac{\partial u}{\partial n} dS dt = \int_D \int_{t_1}^{t_2} k \Delta u dt dx dy dz, \quad (2)$$

由 (1) 与 (2) 得到

$$\int_D \int_{t_1}^{t_2} C \rho \frac{\partial u}{\partial t} - k \Delta u dt dx dy dz = 0,$$

则

$$\begin{cases} C \rho \frac{\partial u}{\partial t} - k \Delta u = 0, & D \times \{t > 0\} \\ u = 0, & D \times \{t = 0\} \\ u(z, y, 0, t)|_{D_1} = u_0, & D_1 = \{(x, y, z) : x^2 + y^2 \leq R^2, z = 0\} \times [0, \infty) \\ k \frac{\partial u}{\partial n}|_{D_2} = \alpha(u_1 - u_2), & D_2 = \{(x, y, z) : x^2 + y^2 = R^2, z \geq 0\} \times [0, \infty) \end{cases}$$

□

1.2 章一第 12 题

任意取定 $y \in M$, 则对任意 $\epsilon \in \mathbb{R}$, 有 $u + \epsilon v \in M$, 记

$$j(\epsilon) = J(u + \epsilon v),$$

由 $J(u) = \min_{y \in M} J(y)$ 知 $j(\epsilon) \geq j(0)$, 从而 $j'(0) = 0$.

由 $J(y)$ 的表达式计算可得

$$j'(\epsilon) = \int_0^1 ((u' + \epsilon y')y' - 2y) dx - y(0).$$

假设 $u \in C^2[0, 1]$, 故当 $\epsilon = 0$ 时, 利用分部积分公式可得

$$\begin{aligned} j'(0) &= \int_0^1 (u'y' - 2y) dx - y(0) \\ &= u'y|_0^1 - \int_0^1 u''y dx - \int_0^1 2y dx - y(0) \\ &= -u'(0)y(0) - \int_0^1 (u''y + 2y) dx - y(0) \\ &= 0. \end{aligned} \quad (3)$$

对 $\forall y \in C_0^\infty[0, 1] \subset M_0$, (3) 式仍成立, 又 $y(1) = y(0) = 0$, 故

$$\int_0^1 (u'' + 2)y dx = 0.$$

因此, 由 $y \in C_0^\infty[0, 1]$ 的任意性可得 $u'' + 2 = 0$, 将此式代入 (3) 中可得

$$u'(0)y(0) - y(0) = 0, \text{ where } y \in M.$$

取 $y(x) = -x + 1$, 则可得

$$u'(0) + 1 = 0,$$

故 u 满足如下边值问题

$$\begin{cases} u'' + 2 = 0 \\ u'(0) + 1 = 0 \\ u(1) = 0 \end{cases} \quad (4)$$

从而解得 $u = -x^2 - x + 2$. 易证 (4) 的解 u 满足 $j'(0) = 0$. 又因为对 $\forall \epsilon \in \mathbb{R}$ 有

$$j''(\epsilon) = \int_0^1 (y')^2 \geq 0,$$

再由 Taylor 公式

$$j(\epsilon) = j(0) + j'(0)\epsilon + \frac{j''(\theta\epsilon)}{2}\epsilon^2, \quad 0 \leq \epsilon \leq 1,$$

可得 $j(0) \leq j(\epsilon)$, 即对 $\forall \epsilon \in \mathbb{R}, \forall y \in M$, 有

$$J(u) \leq J(u + \epsilon y).$$

取 $\epsilon = 1$, 则

$$J(u) \leq J(u + y), \quad \forall y \in M.$$

因此, 对 $\forall v \in M, \exists y \in M$, 使得 $y = v - u$ 且满足

$$J(u) \leq J(u + (v - u)) = J(v),$$

即 $J(u) = \min_{v \in M} J(v)$ 成立。

□

1.3 章一第 14 题

(I) $\Rightarrow$ II

记

$$j(\epsilon) = J(u + \epsilon v),$$

则

$$J(u + \epsilon v) = \frac{1}{2} \int_{\Omega} (|\nabla(u + \epsilon v)|^2 + (u + \epsilon v)^2) dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x)(u + \epsilon v)^2 u v dS - \int_{\Omega} f(u + \epsilon v) dx - \int_{\partial\Omega} g(u + \epsilon v) dS \quad (5)$$

令 $j'(0) = 0$, 则

$$\int_{\Omega} \nabla u \nabla v + uv dx + \int_{\partial\Omega} \alpha(x) uv dS - \int_{\Omega} f v dx - \int_{\partial\Omega} g v dS = 0 \quad (6)$$

则 (6) 与 II 等价。

II $\Rightarrow$ I

将 $\forall w \in M$ 代入 (5) 得到

$$J(w) - J(v) = \frac{1}{2} \int_{\Omega} (|\nabla w|^2 + w^2) - (|\nabla v|^2 + v^2) dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x) (w^2 - v^2) dS - \int_{\Omega} f(w - v) dx - \int_{\partial\Omega} g(w - v) dS$$

令 $w = u + v$, 则

$$J(u+v) - J(u) = \frac{1}{2} \int_{\Omega} (|\nabla v|^2 + v^2) dx + \frac{1}{2} \int_{\partial\Omega} \alpha(x) v^2 dS - \int_{\Omega} (\nabla u \nabla v + uv - fv) dx + \int_{\partial\Omega} (\alpha(x) uv - gv) dS \geq 0$$

故满足 II 中的 u 使得 $J(u)$ 为最小值。

(2) 只需证明 II $\Leftrightarrow$ III

设 $u \in C^2(\bar{\Omega})$, 则

II $\Rightarrow$ III

$$\begin{aligned} & \int_{\Omega} (\nabla u \nabla v + uv - fv) dx + \int_{\partial\Omega} (\alpha(x) uv - gv) dS \\ &= \int_{\partial\Omega} v \frac{\partial u}{\partial n} dS - \int_{\Omega} v \Delta u dx + \int_{\Omega} uv - fv dx + \int_{\partial\Omega} \alpha(x) uv - gv dS \\ &= \int_{\Omega} (-\Delta u + u - f) v dx + \int_{\partial\Omega} v (\alpha(x) u + \frac{\partial u}{\partial n} - g) dS \\ &= 0 \end{aligned} \quad (7)$$

又由 v 的任意性, 取 $v_1 \in C_0^\infty(\Omega)$, 则有 $v_1|_{\partial\Omega} = 0$, 代入 (5) 由引理 2.1 得到

$$-\Delta u + u - f = 0, \quad x \in \Omega$$

同理取 $v_2 \in C_0^\infty(\partial\Omega)$, 则有 $v_2|_{\Omega} = 0$, 代入 (5) 由引理 2.1 得到

$$\alpha(x) u + \frac{\partial u}{\partial n} - g = 0, \quad x \in \partial\Omega$$

III $\Rightarrow$ II

因为 $u \in C^2(\bar{\Omega})$, 由分部积分得到

$$\int_{\Omega} (\nabla u \nabla v + uv - fv) dx + \int_{\partial\Omega} (\alpha(x) uv - gv) dS = \int_{\Omega} (-\Delta u + u - f) v dx + \int_{\partial\Omega} v (\alpha(x) u + \frac{\partial u}{\partial n} - g) dS = 0$$

□

1.4 章一第 16 题

(1) 直接进行计算,

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}, \\ \frac{\partial u}{\partial t} &= -a \frac{\partial u}{\partial \xi} + a \frac{\partial u}{\partial \eta}, \\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \eta^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta}, \\ \frac{\partial^2 u}{\partial t^2} &= a^2 \frac{\partial^2 u}{\partial \xi^2} + a^2 \frac{\partial^2 u}{\partial \eta^2} - 2a^2 \frac{\partial^2 u}{\partial \xi \partial \eta}.\end{aligned}\quad (8)$$

将 (8) 代入 $u_{tt} - a^2 u_{xx} = 0$ 得到 $u_{\xi\eta} = 0$ 。先对 η 积分再对 ξ 积分得到

$$u_{\xi} = f(\xi)$$

$$u = \int f(\xi) d\xi + g(\eta) := F(\xi) + G(\eta) = F(x - at) + G(x + at)$$

其中 $f(\xi)$, $g(\eta)$ 分别是只与 ξ , η 有关的函数。

(2) 直接进行计算,

$$u_t = -au_{\xi} + u_{\tau}, \quad u_x = u_{\xi}, \quad u_{xx} = u_{\xi\xi}, \quad (9)$$

将 (9) 代入 $u_t + au_x = a^2 u_{xx}$ 得

$$u_{\tau} = a^2 u_{\xi\xi}$$

□

1.5 章一第 18 题

直接计算并代入

$$u_t = -\frac{x}{2t^{3/2}} \tilde{u}', \quad u_x = -\frac{1}{\sqrt{t}} \tilde{u}', \quad u_{xx} = \frac{1}{t} \tilde{u}''.\quad (10)$$

将 (10) 代入 $u_t - a^2 u_{xx} = 0$ 得

$$\frac{x}{2t^{3/2}} \tilde{u}' + a^2 \frac{1}{t} \tilde{u}'' = 0 \Rightarrow \tilde{u}' + \frac{2a^2}{\xi} \tilde{u}'' = 0$$

则定解问题转化为

$$\begin{cases} \tilde{u}' + \frac{2a^2}{\xi} \tilde{u}'' = 0, & 0 < \xi < \infty \\ \tilde{u}|_{\xi=0} = 0, \\ \tilde{u}|_{\xi=\infty} = u_0 \end{cases}$$

解上述常微分方程组得到

$$u(x, t) = \frac{u_0}{a\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{t}}} e^{-\frac{\eta^2}{4a^2}} d\eta.$$

□

二. PDE HOMEWORK 2

2.1 章二第 3 题

(1) 求特征线, 特征方程

$$\begin{cases} \frac{dx}{dt} = 2, \\ x(0) = c. \end{cases}$$

的解为

$$x(t) = 2t + c. \quad (11)$$

令 $U(t) = u(x(t), t)$, 代入原方程得:

$$\begin{cases} \frac{dU(t)}{dt} = 0, \\ U(0) = c^2. \end{cases}$$

解得

$$U(t) = c^2.$$

又由 (11) 得 $c = x - 2t$, 代入得

$$u(x, t) = (x - 2t)^2.$$

(2) 求特征线, 特征方程

$$\begin{cases} \frac{dx}{dt} = 2, \\ x(0) = c. \end{cases}$$

的解为

$$x(t) = 2t + c. \quad (12)$$

令 $U(t) = u(x(t), t)$, 代入原方程得:

$$\begin{cases} \frac{dU(t)}{dt} + U(t) = (2t + c)t, \\ U(0) = 2 - c. \end{cases}$$

解得

$$U(t) = -2e^{-t} + 2t^2 + (c - 4)t + 4 - c.$$

又由 (12) 得 $c = x - 2t$, 代入得

$$\begin{aligned} u(x, t) &= -2e^{-t} + 2t^2 + (x - 2t - 4)(t - 1) \\ &= -2e^{-t} + xt - x - 2t + 4. \end{aligned}$$

(3) 求特征线, 特征方程

$$\begin{cases} \frac{dx}{dt} = -\frac{1}{2}, \\ x(0) = c. \end{cases}$$

的解为

$$x(t) = -\frac{1}{2}t + c. \quad (13)$$

令 $U(t) = u(x(t), t)$, 代入原方程得

$$\begin{cases} \frac{dU(t)}{dt} + (\frac{1}{4}t + \frac{c}{2})U(t) = 0, \\ U(0) = 2ce^{\frac{c^2}{2}}. \end{cases}$$

解得

$$U(t) = 2ce^{\frac{1}{8}t^2 - \frac{c}{2}t + \frac{c^2}{2}}.$$

又由 (13) 得 $c = x + \frac{1}{2}t$, 代入得

$$u(x, t) = (2x + t)e^{\frac{x^2}{2}}.$$

(4) 求特征线, 特征方程

$$\begin{cases} \frac{dx}{dt} = 1 + x^2, \\ x(0) = c. \end{cases}$$

的解为

$$x(t) = \tan(t + \arctan c). \quad (14)$$

令 $U(t) = u(x(t), t)$, 代入原方程得

$$\begin{cases} \frac{dU(t)}{dt} - U(t) = 0, \\ U(0) = \arctan c. \end{cases}$$

解得

$$U(t) = e^t \arctan c.$$

又由 (14) 得 $c = \tan(\arctan x - t)$, 则

$$u(x, t) = e^t (\arctan x - t).$$

□

2.2 章二第 4 题

原方程

$$\begin{aligned} (1 - \frac{x}{h})^2 \frac{\partial^2 u}{\partial t^2} &= a^2 \frac{\partial}{\partial x} ((1 - \frac{x}{h})^2 \frac{\partial u}{\partial x}), \\ (1 - \frac{x}{h})^2 \frac{\partial^2 u}{\partial t^2} &= a^2 (2(-\frac{h-x}{h^2}) \frac{\partial u}{\partial x} + (1 - \frac{x}{h})^2 \frac{\partial^2 u}{\partial x^2}), \\ (h-x) \frac{\partial^2 u}{\partial t^2} &= a^2 (-2 \frac{\partial u}{\partial x} + (h-x) \frac{\partial^2 u}{\partial x^2}), \\ \frac{\partial^2 (h-x)u}{\partial t^2} &= a^2 \frac{\partial^2 (h-x)u}{\partial x^2}. \end{aligned}$$

令 $v = (h - x)u$, 则有

$$\frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2}.$$

由第一章习题 16.(1) 得到方程的解为

$$v(x, t) = F(x - at) + G(x + at).$$

从而原方程的通解可以表为

$$u(x, t) = \frac{F(x - at) + G(x + at)}{h - x}.$$

□

2.3 章二第 5 题

设方程的解为

$$u(x, t) = F(x + at) + G(x - at),$$

若方程只有右行波, 则方程的解为

$$u(x, t) = G(x - at). \quad (15)$$

又由初值条件

$$u(x, 0) = G(x) = \varphi(x),$$

对 (15) 方程两边同时对 t 求导, 且令 $t = 0$ 得

$$-aG'(x) = \psi(x).$$

从而方程只有右行波 $\Leftrightarrow -a\varphi'(x) = \psi(x)$.

□

2.4 章二第 6 题

设方程的解为

$$u(x, t) = F(x + at) + G(x - at),$$

代入可得

$$F(x + ax) + G(x - ax) = u_0(x), \quad (16)$$

$$F'(x + ax) - G'(x - ax) = u_1(x). \quad (17)$$

对 (17) 在 $[0, x]$ 上积分得

$$\frac{1}{1+a}F(x+ax) - \frac{1}{1-a}G(x-ax) = \int_0^x u_1(\xi)d\xi + c. \quad (18)$$

联立 (16) 与 (18) 可得

$$\begin{cases} F(x+ax) = \frac{1+a}{2}u_0(x) + \frac{1-a^2}{2}\int_0^x u_1(\xi)d\xi + c\frac{1-a^2}{2}, \\ G(x-ax) = \frac{1-a}{2}u_0(x) - \frac{1-a^2}{2}\int_0^x u_1(\xi)d\xi - c\frac{1-a^2}{2}, \end{cases}$$

$\Rightarrow$

$$\begin{cases} F(x) = \frac{1+a}{2}u_0(\frac{x}{1+a}) + \frac{1-a^2}{2}\int_0^{\frac{x}{1+a}} u_1(\xi)d\xi + c\frac{1-a^2}{2}, \\ G(x) = \frac{1-a}{2}u_0(\frac{x}{1-a}) - \frac{1-a^2}{2}\int_0^{\frac{x}{1-a}} u_1(\xi)d\xi - c\frac{1-a^2}{2}, \end{cases}$$

从而

$$u(x, t) = \frac{1}{2}((1+a)u_0(\frac{x+t}{1+a}) + (1-a)u_0(\frac{x-t}{1-a})) + \frac{1-a^2}{2}\int_{\frac{x-t}{1-a}}^{\frac{x+t}{1+a}} u_1(\xi)d\xi.$$

由解的表达式可知对任意满足 $t > ax$ 的点 (x, t) 的依赖区间为

当 $a^2 < 1$ 时, 依赖区间 $[\frac{x-t}{1-a}, \frac{x+t}{1+a}]$,

当 $a^2 > 1$ 时, 依赖区间 $[\frac{x+t}{1+a}, \frac{x-t}{1-a}]$.

由于初值只给定在 $c \leq x \leq b$ 上, 要确定解, 需要依赖区间落在 $[c, b]$ 中。下面讨论确定解的区域, 不妨设 $0 \leq c \leq b$,

Case 1 $a^2 < 1$ ($-1 < a < 1$).

要满足 $[\frac{x-t}{1-a}, \frac{x+t}{1+a}] \subseteq [c, b]$, 即

$$\begin{cases} t > ax, \\ t \leq x - c(1-a), \\ t \leq b(1+a) - x. \end{cases}$$

Case 2 $a^2 > 1$ ($a < -1$ or $a > 1$).

要满足 $[\frac{x-t}{1-a}, \frac{x+t}{1+a}] \subseteq [c, b]$, 即

当 $a < -1$ 时

$$\begin{cases} t > ax, \\ t \leq -x + c(a+1), \\ t \geq x + b(a-1). \end{cases}$$

当 $a > 1$ 时

$$\begin{cases} t > ax, \\ t \geq -x + c(a+1), \\ t \leq x + b(a-1). \end{cases}$$

□

2.5 章二第 7 题

(1) ” $\Leftarrow$ ” 若方程组有解, 设解为

$$u(x, t) = F(x+t) + G(x-t) + (t^3 - x^3),$$

则

$$F(2x) + G(0) = 0 \Rightarrow F(2x) = \text{const}, F'(2x) = 0.$$

又代入原方程得

$$F'(2x) - G'(0) + 3x^2 = u_1(x) \Rightarrow u_1(x) - 3x^2 = \text{const}.$$

又因为 F 与 G 只要满足上述条件都是方程的解, 而 F 与 G 不唯一, 因此解不唯一。

” $\Rightarrow$ ”

若 $u_1(x) - 3x^2 = \text{const}$, 取特解

$$u(x, t) = t^3 - x^3 - c(x - t),$$

则 u 满足

$$\begin{cases} u_{tt} - u_{xx} = 6(x + t), \\ u|_{t=x} = 0, \\ u_t|_{t=x} = u_1(x). \end{cases}$$

且有 $u_1(x) - 3x^2 = c$.

(2) 若初值给定在直线 $t = ax$, $a = \pm 1$ 上, 只讨论 $a = +1$ 的情况, $a = -1$ 的情况同理可得。假设原问题有两个解, 记为 $\tilde{u}_1, \tilde{u}_2$, 则 $u := \tilde{u}_1 - \tilde{u}_2$ 满足方程组

$$\begin{cases} u_{tt} - u_{xx} = 0, \\ u|_{t=x} = 0, \\ u_t|_{t=x} = 0. \end{cases}$$

该方程的通解为 $u = f(x - t) + g(x + t)$, 代入初值条件可得

$$\begin{cases} f(0) + g(2x) = 0, \\ -f'(0) + g'(2x) = 0. \end{cases}$$

解得 $g(2x) = -f(0) = 2f'(0)x + c$, 从而

$$c = -f(0), f'(0) = 0.$$

此时选取满足 $f'(0) = 0$ 的 u 即为解, 如 $u = (x + t)^2$, 此时 u 不恒为 0, 且原 Cauchy 问题的解不唯一。

若初值给定在直线 $t = ax$, $a \neq \pm 1$ 上, 假设原问题有两个解, 记为 $\tilde{u}_1, \tilde{u}_2$, 则 $u := \tilde{u}_1 - \tilde{u}_2$ 满足方程组

$$\begin{cases} u_{tt} - u_{xx} = 0, \\ u|_{t=ax} = 0, \\ u_t|_{t=ax} = u_1(x). \end{cases}$$

该方程的通解为 $u = f(x - t) + g(x + t)$, 代入初值条件可得

$$\begin{cases} f((1 - a)x) + g((1 + a)x) = 0, \\ -f'((1 - a)x) + g'((1 + a)x) = 0. \end{cases}$$

解得 $f((1-a)x) = -\frac{c}{2}(1-a^2)$, $g((1+a)x) = \frac{c}{2}(1-a^2)$, 从而

$$u = f(x-t) + g(x+t) = 0.$$

因此原问题的解唯一。

□

2.6 章二第 17 题

设方程的解为 $u(x, t) = F(x+t) + G(x-t)$, 当 $-\infty < x \leq 0$ 时, 令 $t = -x$, 则

$$F(0) + G(2x) = \varphi(x) \Rightarrow G(x) = \varphi\left(\frac{x}{2}\right) - F(0).$$

当 $0 \leq x < \infty$ 时, 令 $t = x$, 则

$$F(2x) + G(0) = \psi(x) \Rightarrow F(x) = \psi\left(\frac{x}{2}\right) - G(0).$$

从而当 $t \geq |x|$ 时,

$$\begin{aligned} u(x, t) &= \psi\left(\frac{x+t}{2}\right) + \varphi\left(\frac{x-t}{2}\right) - (F(0) + G(0)) \\ &= \psi\left(\frac{x+t}{2}\right) + \varphi\left(\frac{x-t}{2}\right) - \psi(0). \end{aligned}$$

又 $\varphi(x)$ 给定在 $(-a, 0]$, $\psi(x)$ 给定在 $[0, b]$, 故定解条件的决定区域为

$$\begin{cases} x+t \geq 0, \\ x+t \leq 2b, \\ x-t \leq 0, \\ x-t \geq 2a. \end{cases}$$

□

2.7 章二第 30 题

(1)

$$\begin{cases} u_t + au_x = 0, & x > 0, t > 0, a > 0, \\ u(0, t) = \mu(t), & t \geq 0, \\ u(x, 0) = \varphi(x), & x \geq 0. \end{cases}$$

该初边值问题的提法正确。

令 $U(t) = u(x(t), t)$, 代入原方程可得

$$\begin{cases} \frac{dU(t)}{dt} = 0, \\ U(0) = \varphi(c). \end{cases}$$

得

$$u(t) = \varphi(c).$$

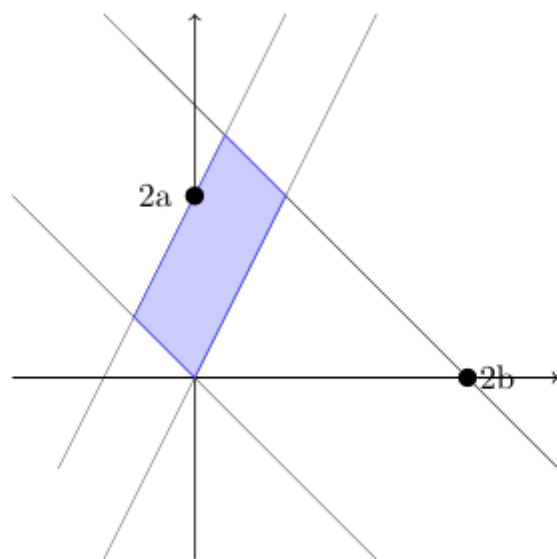


图 1: pic1

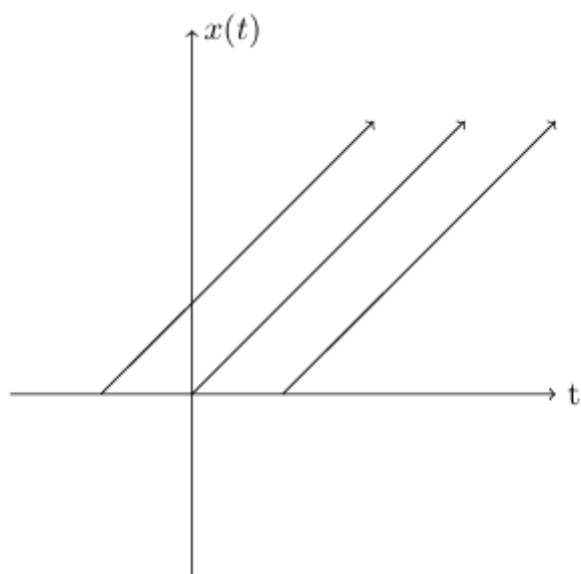


图 2: pic2

由 $x = at + c$, $c = x - at$ 得

$$u(x, t) = \varphi(x - at), (x, t) \in \{0 < at < x\}.$$

注意到 $u_x + \frac{1}{a}u_t = 0$, 同理令 $V(x) = u(x, t(x))$, 代入原方程得

$$\begin{cases} \frac{dV(x)}{dx} = 0, \\ V(0) = \mu(d). \end{cases}$$

由方程 $t = \frac{1}{a}x + d$ 得

$$u(x, t) = \mu(t - \frac{1}{a}x), (x, t) \in \{0 < x < at\}.$$

若解 u 是古典解, 即

$$u \in C^1((0, \infty) \times (0, \infty)) \cap C([0, \infty) \times [0, \infty)).$$

又因为 $C([0, \infty) \times [0, \infty))$, 则在 $\{x = at\}$ 上, $\varphi(x - at) = \mu(t - \frac{1}{a}x)$, 即

$$\varphi(0) = \mu(0).$$

由 $u \in C^1((0, \infty) \times (0, \infty))$, 得 $\mu \in C^1([0, \infty))$, $\varphi \in C^1([0, \infty))$, 且

$$\begin{cases} \varphi'(0) = -\frac{1}{a}\mu'(0), \\ -a\varphi'(0) = \mu'(0). \end{cases}$$

即

$$a\varphi'(0) + \mu'(0) = 0.$$

(2) 对于方程组

$$\begin{cases} u_t - au_x = 0, & x > 0, t > 0, a > 0, \\ u(0, t) = \mu(t), & t \geq 0, \\ u(x, 0) = \varphi(x), & x \geq 0. \end{cases}$$

该方程的特征方程为

$$\begin{cases} \frac{dx}{dt} = -a, \\ x(0) = c. \end{cases}$$

解得

$$x(t) = c - at.$$

在 t 得正半轴上任取一点作与特征线平行的直线, 当特征线方程为 $x = -at + c$, ($a > 0$) 时, 该直线会与 x 轴相交, 从而 t 的值可以被 x 轴上对应的点确定, 此时 t 不能任意取值。而当特征线方程为 $x = -at + c$, ($a > 0$) 时, 任取 t 正半轴上一点, 过该点作与特征线平行的直线, 其方向总是保持向上。因此, 方程 $u_t + au_x = 0$ 的初边值问题提法正确。

□

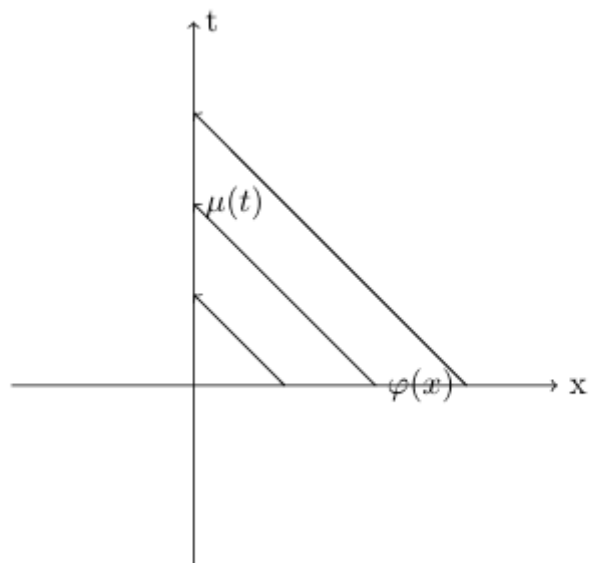


图 3: pic3

2.8 章二第 31 题

令 $v = x + u - 1$, 则方程化为

$$\begin{cases} v_t = -\frac{1}{v}v_x, & x \in \mathbb{R}, t > 0, \\ v(x, 0) = x - 1, & x \in \mathbb{R}. \end{cases}$$

令 $w(t) = v(x(t), t)$, 则

$$\begin{cases} x'(t) = \frac{1}{v(x(t), t)} = \frac{1}{w}, \\ w'(0) = 0, \\ x(0) = c, w(0) = c - 1. \end{cases}$$

解得

$$x = \frac{x}{c-1} + c.$$

给定 (x, t) 找 c 使得特征线通过 (x, t) , 则 $c^2 - (x+1)c + x+t = 0$, 求解得

$$c = \frac{x+1 \pm \sqrt{(x-1)^2 - 4t}}{2}.$$

当 $(x-1)^2 - 4t > 0$ 时有解,

$$\begin{aligned} w = c - 1 &= \frac{x-1 \pm \sqrt{(x-1)^2 - 4t}}{2} \\ v(x, t) &= \begin{cases} \frac{x-1 + \sqrt{(x-1)^2 - 4t}}{2}, & x > 1, \\ \frac{x-1 - \sqrt{(x-1)^2 - 4t}}{2}, & x < 1, \end{cases} \\ u(x, t) &= \begin{cases} \frac{-x+1 + \sqrt{(x-1)^2 - 4t}}{2}, & x > 1, \\ \frac{-x+1 - \sqrt{(x-1)^2 - 4t}}{2}, & x < 1, \end{cases} \end{aligned}$$

□

三. PDE HOMEWORK 3

3.1 章二第 8 题

将原方程拆分为三个方程

$$\begin{cases} U_{tt} - a^2 U_{xx} = 0, \\ U|_{t=0} = f(x), \\ U_t|_{t=0} = 0. \end{cases}$$

$$\begin{cases} V_{tt} - a^2 V_{yy} = 0, \\ V|_{t=0} = g(y), \\ V_t|_{t=0} = \varphi(y). \end{cases}$$

$$\begin{cases} W_{tt} - a^2 W_{zz} = 0, \\ W|_{t=0} = 0, \\ W_t|_{t=0} = \psi(z). \end{cases}$$

由 D'Alembert 公式得

$$\begin{aligned} U(x, t) &= \frac{1}{2}(f(x+at) + f(x-at)), \\ V(y, t) &= \frac{1}{2}(g(y+at) + g(y-at)) + \frac{1}{2a} \int_{y-at}^{y+at} \varphi(\xi) d\xi, \\ W(z, t) &= \frac{1}{2a} \int_{z-at}^{z+at} \psi(\xi) d\xi, \end{aligned}$$

原方程的解为

$$u(x, y, z, t) = U(x, t) + V(y, t) + W(z, t).$$

□

3.2 章二第 13 题

使用能量不等式的方法，对方程移项，两边同乘 u_t 并在 K_τ 上积分得到

$$\iint_{K_\tau} (u_{tt} - a^2 u_{xx}) u_t dt dx = \iint_{K_\tau} (f(x, t) - b(x, t) u_x - c(x, t) u_t) u_t dx dt. \quad (19)$$

注意到

$$(u_{tt} - a^2 u_{xx}) u_t = \frac{1}{2} (u_t^2)_t + \frac{1}{2} a^2 (u_x^2)_t - (a^2 u_t u_x)_x,$$

由 Green 公式，则 (19) 的左端

$$\begin{aligned} LHS &= \left(\int_{\Omega_\tau} + \int_{\Omega_0} + \int_{\Gamma_\tau^1 \cup \Gamma_\tau^2} \right) (a^2 u_t u_x dt + \frac{1}{2} (u_t^2 + a^2 u_x^2) dx) \\ &:= J_1 + J_2 + J_3, \end{aligned}$$

其中 Ω_τ , Ω_0 , Γ_τ^1 , Γ_τ^2 的定义同书 p45 图 2.5。

由于在 Γ_τ^1 上 $dx = a dt$, 在 Γ_τ^2 上 $dx = -a dt$, 代入初值条件得

$$\begin{aligned} J_1 &= \int_{\Omega_\tau} \frac{1}{2} (u_t^2(x, \tau) + a^2 u_x^2(x, \tau)) dx, \\ J_2 &= - \int_{\Omega_0} \frac{1}{2} (\psi^2(x) + a^2 \psi_x^2(x)) dx, \\ J_3 &= -\frac{1}{2} a \int_{\Gamma_\tau^1} \left(\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} \right)^2 dt + \frac{1}{2} a \int_{\Gamma_\tau^2} \left(\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} \right)^2 dt. \end{aligned}$$

沿着 K_τ 逆时针方向在 Γ_τ^1 上 dt 为负, 在 Γ_τ^2 上 dt 为正, 因此

$$J_3 \geq 0.$$

对 (19) 的右端

$$\begin{aligned} RHS &= \iint_{K_\tau} (f(x, t) - b(x, t)u_x - c(x, t)u_t)u_t dx dt \\ &\leq \frac{1}{2} \iint_{K_\tau} f^2 dx dt + \frac{3}{2} \iint_{K_\tau} u_t^2 dx dt + \frac{\tilde{b}^2}{2} \iint_{K_\tau} u_x^2 dx dt + \frac{\tilde{c}^2}{2} \iint_{K_\tau} u_t^2 dx dt \\ &= \frac{1}{2} \iint_{K_\tau} f^2 dx dt + \frac{\tilde{b}^2}{2} \iint_{K_\tau} u_x^2 dx dt + \frac{\tilde{c}^2 + 3}{2} \iint_{K_\tau} u_t^2 dx dt \\ &\leq \frac{1}{2} \iint_{K_\tau} f^2 dx dt + \frac{M}{2} \iint_{K_\tau} u_t^2 + a^2 u_x^2 dx dt \end{aligned}$$

其中 $\tilde{b} = (\max_{K_\tau} |b(x, t)|)^2$, $\tilde{c} = (\max_{K_\tau} |c(x, t)|)^2$, $M = \max_{K_\tau} \left\{ \frac{\tilde{b}^2}{a^2}, 3 + \tilde{c}^2 \right\}$ 。

则

$$\begin{aligned} LHS &\geq \int_{\Omega_\tau} \frac{1}{2} (u_t^2(x, \tau) + a^2 u_x^2(x, \tau)) dx - \int_{\Omega_0} \frac{1}{2} (\psi^2(x) + a^2 \psi_x^2(x)) dx, \\ RHS &\leq \frac{1}{2} \iint_{K_\tau} f^2 dx dt + \frac{M}{2} \iint_{K_\tau} u_t^2 + a^2 u_x^2 dx dt. \end{aligned}$$

$$\Rightarrow \int_{\Omega_\tau} (u_t^2(x, \tau) + a^2 u_x^2(x, \tau)) dx - \int_{\Omega_0} (\psi^2(x) + a^2 \psi_x^2(x)) dx \leq \iint_{K_\tau} f^2 dx dt + M \iint_{K_\tau} u_t^2 + a^2 u_x^2 dx dt.$$

由 Gronwall 不等式与 τ 的任意性得

$$\begin{aligned} \int_{\Omega_\tau} (u_t^2(x, \tau) + a^2 u_x^2(x, \tau)) dx &\leq M_2 \left(\int_{\Omega_0} (\psi^2(x) + a^2 \psi_x^2(x)) dx + \iint_{K_\tau} f^2 dx dt \right) \\ &\Rightarrow \int_{K_\tau} (u_t^2 + a^2 u_x^2) dx \leq M_1 \left(\int_{\Omega_0} (\psi^2(x) + a^2 \psi_x^2(x)) dx + \iint_{K_\tau} f^2 dx dt \right) \end{aligned}$$

若原问题存在两个解 u_1, u_2 , 令 $\tilde{u} = u_1 - u_2$ 为 $f = \varphi = \psi = 0$ 时方程得解。由上述能量估计知

$$\tilde{u} = 0.$$

即原问题的解是唯一的。

□

3.3 章二第 14 题

由 Poisson 公式知, 解

$$u(x, y, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \left(\int_{B_{at}(x, y)} \frac{\varphi(z)}{\sqrt{a^2 t^2 - (z_2 - y)^2 - (z_1 - x)^2}} dz \right) + \frac{1}{2\pi a} \int_{B_{at}(x, y)} \frac{\psi(z)}{\sqrt{a^2 t^2 - (z_2 - y)^2 - (z_1 - x)^2}} dz.$$

解得依赖区域为以 (x, y) 为中心, at 为半径的圆 $B_{at}(x, y)$ 。又

$$\varphi(x, y), \psi(x, y) = \begin{cases} 0, & (x, y) \in \Omega, \\ \text{positive number}, & (x, y) \in \mathbb{R}^2 - \Omega. \end{cases}$$

所以当且仅当 $B_{at}(x, y) \in \Omega$ 时, 有 $u = 0$ 。因此, 要求的区域需满足

$$\begin{cases} -1 \leq x - at \leq 1, \\ -1 \leq x + at \leq 1, \\ -1 \leq y - at \leq 1, \\ -1 \leq y + at \leq 1, \\ t \geq 0. \end{cases}$$

即为一个以 Ω 为底, 高为 $\frac{1}{a}$ 的正四棱锥。

□

3.4 章二第 21 题

由 Poisson 公式,

$$u(x, y, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \left(\int_{B_{at}(x)} \frac{\varphi(z)}{\sqrt{a^2 t^2 - (z_2 - x_2)^2 - (z_1 - x_1)^2}} dz \right) + \frac{1}{2\pi a} \int_{B_{at}(x)} \frac{\psi(z)}{\sqrt{a^2 t^2 - (z_2 - x_2)^2 - (z_1 - x_1)^2}} dz + \frac{1}{2\pi a} \int_0^t \int_{B_{a(t-\tau)}(x)} \frac{f(\tau, z)}{\sqrt{a^2 (t-\tau)^2 - (z_2 - x_2)^2 - (z_1 - x_1)^2}} dz d\tau.$$

知该 Cauchy 问题解的表达式为

$$u(x, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \left(\int_{(\xi-x)^2 + (\eta-y)^2 \leq (at)^2} \frac{\xi^2(\xi+\eta)}{\sqrt{a^2 t^2 - (\xi-x)^2 - (\eta-y)^2}} d\xi d\eta \right).$$

令 $I = \int_{(\xi-x)^2 + (\eta-y)^2 \leq (at)^2} \frac{\xi^2(\xi+\eta)}{\sqrt{a^2 t^2 - (\xi-x)^2 - (\eta-y)^2}} d\xi d\eta$, 则由极坐标变换 $\xi = x + r\cos\theta$, $\eta = y + r\sin\theta$ 可得

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^{at} \frac{(x + r\cos\theta)^2(x + y + r\cos\theta + r\sin\theta)}{\sqrt{a^2 t^2 - r^2}} r dr d\theta \\ &= \int_0^{at} \frac{2\pi x^2(x + y) + \pi r^2(x + y) + 2\pi r^2 x}{\sqrt{a^2 t^2 - r^2}} r dr \\ &= 2a\pi x^2(x + y)t + \frac{2}{3}a^3 t^3(\pi(x + y) + 2\pi x). \end{aligned}$$

从而

$$\begin{aligned}\frac{1}{2\pi a} \frac{\partial}{\partial t} I &= \frac{1}{2\pi a} (2a\pi x^2(x+y) + 2a^3 t^2(3\pi x + \pi y)) \\ &= x^2(x+y) + a^2 t^2(3x+y).\end{aligned}$$

综上, 解得

$$u(x, y, t) = x^2(x+y) + a^2 t^2(3x+y).$$

□

3.5 章二第 23 题

(1) 设 $u(x, t) = X(x)T(t) \neq 0$, 代入方程有

$$\begin{aligned}X(x)T''(t) - a^2 X''(x)T(t) &= 0, \\ \Leftrightarrow \frac{T''(t)}{a^2 T(t)} &= \frac{X''(x)}{X(x)} = -\lambda.\end{aligned}$$

从而

$$\begin{cases} T''(t) + a^2 \lambda T(t) = 0, \\ X''(x) + \lambda X(x) = 0. \end{cases}$$

由边界条件知 $X(0) = X(l) = 0$, 解得

$$X(x) = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x, \quad \lambda_n = \left(\frac{n\pi}{l}\right)^2.$$

则

$$X_n(x) = C_n \sin \frac{n\pi}{l} x,$$

又

$$T_n = a_n \cos \frac{n\pi a}{l} t + b_n \sin \frac{n\pi a}{l} t,$$

则

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos \frac{an\pi}{l} t + B_n \sin \frac{an\pi}{l} t) \sin \frac{n\pi}{l} x, \quad (20)$$

代入初值条件得

$$\begin{cases} u(x, 0) = \sin^2 \frac{\pi}{l} x = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l} x, \\ u_t(x, 0) = x(l-x) = \sum_{n=1}^{\infty} B_n \frac{an\pi}{l} \sin \frac{n\pi}{l} x. \end{cases}$$

将 $\sin^2 \frac{\pi}{l} x$, $x(l-x)$ 都按照 $\{\sin \frac{n\pi}{l} x\}$ 展开, 对比系数得到

$$A_n = \frac{2}{l} \int_0^l \sin \frac{n\pi x}{l} \sin^2 \frac{\pi x}{l} dx,$$

$$B_n = \frac{2}{an\pi} \int_0^l \sin \frac{n\pi x}{l} x(l-x) dx.$$

代入 (20) 即为所求。

(2) 与 (1) 完全同理得

$$A_n = \frac{2}{l} \int_0^l \sin \frac{n\pi x}{l} x(l-x) dx,$$

$$B_n = 0.$$

代入 (20) 即为所求。

(3) 设 $u(x, t) = X(x)T(t) \neq 0$, 代入得

$$XT'' - a^2 X''T + XT = 0.$$

设 $\frac{X''}{X} = -\lambda$, 代入得

$$T'' + (\lambda a^2 + 1)T = 0.$$

则 $X = c_1 \cos \sqrt{\lambda}x + c_2 \sin \sqrt{\lambda}x$, 由 $u'(0, t) = 0$ 知 $c_2 = 0$, 取 $\lambda = \frac{n^2\pi^2}{l^2}$, 得

$$X_n(x) = C_n \cos \frac{n\pi}{l}x,$$

$$T_n(t) = a_n \cos \sqrt{\lambda_n a^2 + 1}t + b_n \sin \sqrt{\lambda_n a^2 + 1}t.$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} X_n T_n = \sum_{n=1}^{\infty} (A_n \cos \sqrt{\lambda_n a^2 + 1}t + B_n \sin \sqrt{\lambda_n a^2 + 1}t) \cos \frac{n\pi}{l}x.$$

代入初值条件得

$$\begin{cases} u(x, 0) = \varphi(x) = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{l}x, \\ u_t(x, 0) = \psi(x) = \sum_{n=1}^{\infty} B_n \sqrt{\lambda_n a^2 + 1} \cos \frac{n\pi}{l}x. \end{cases}$$

将 φ, ψ 都按照 $\{\cos \frac{n\pi}{l}x\}$ 展开, 对比系数得到

$$\begin{cases} \frac{2}{l} \int_0^l \varphi(x) \cos \frac{n\pi}{l}x dx = A_n, \\ \frac{2}{l} \int_0^l \psi(x) \cos \frac{n\pi}{l}x dx = \sqrt{\lambda_n a^2 + 1} B_n. \end{cases} \quad (21)$$

(a) 将 $\varphi(x) = \psi(x) = 1$ 代入 (21) 得

$$A_n = \begin{cases} 0, & n \geq 1, \\ 1, & n = 0. \end{cases} \quad B_n = \begin{cases} 0, & n \geq 1, \\ 1, & n = 0. \end{cases}$$

$$\Rightarrow u(x, t) = \cos t + \sin t.$$

(b) 将 $\varphi(x) = \cos \frac{\pi x}{l}$, $\psi(x) = 0$ 代入 (21) 得

$$A_n = \begin{cases} 0, & n \neq 1, \\ 1, & n = 1. \end{cases} \quad B_n = 0.$$

$$\Rightarrow u(x, t) = \cos \sqrt{\frac{\pi^2}{l^2} a^2 + 1}t \cos \frac{\pi}{l}x.$$

□

3.6 章二第 26 题

(3) 令 $u(x, t) = v(x, t) + \frac{B}{l}x$, 则 $v(x, t)$ 满足定解问题

$$\begin{cases} v_{tt} - a^2 v_{xx} = A, & (x, t) \in Q, \\ v|_{t=0} = 0, \quad v_t|_{t=0} = 0, & 0 \leq x \leq l, \\ v_x|_{x=0} = 0, \quad v_x|_{x=l} = 0, & t \geq 0. \end{cases} \quad (22)$$

设 $v(x, t) = \sum_{n=1}^{\infty} X_n(x)T_n(t)$, 由

$$\begin{cases} X''(x) + \lambda X(x) = 0, \\ X(0) = X(l) = 0. \end{cases}$$

解得 $\lambda_n = (\frac{n\pi}{l})^2$, $X_n(x) = \sin \frac{n\pi}{l}x$, 代入 (22) 可得

$$\begin{cases} \sum_{n=1}^{\infty} (T_n'' + (\frac{an\pi}{l})^2 T_n) \sin \frac{n\pi x}{l} = A, \\ \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l} |_{t=0} = 0, \\ \sum_{n=1}^{\infty} T_n'(t) \sin \frac{n\pi x}{l} |_{t=0} = 0. \end{cases} \quad (23)$$

全体特征函数 $\{\sin \frac{n\pi x}{l}\}$ 组成了 $L^2[0, l]$ 上的完备正交系, 因此由傅里叶系数得唯一性可得

$$\begin{cases} T_n'' + (\frac{an\pi}{l})^2 T_n = \frac{2}{l} \int_0^l a \sin \frac{n\pi x}{l} dx = A \frac{2}{n\pi} (1 - \cos n\pi), \\ T_n(0) = T_n'(0) = 0. \end{cases} \quad (24)$$

当 n 为偶数时, 解得 $T_n(t) = 0$ 。当 n 为奇数时, 解得 $T_n(t) = \frac{4Al^2}{a^2 n^3 \pi^3} (1 - \cos \frac{n\pi at}{l})$, 因此

$$v(x, t) = \sum_{k=0}^{\infty} \frac{4Al^2}{a^2 (2k+1)^3 \pi^3} (1 - \cos \frac{(2k+1)\pi at}{l}) \sin \frac{(2k+1)\pi x}{l}.$$

得

$$\begin{aligned} u(x, t) &= v(x, t) + \frac{B}{l}x \\ &= \sum_{k=0}^{\infty} \frac{4Al^2}{a^2 (2k+1)^3 \pi^3} (1 - \cos \frac{(2k+1)\pi at}{l}) \sin \frac{(2k+1)\pi x}{l} + \frac{B}{l}x. \end{aligned}$$

□

3.7 章二第 27 题

该定解问题对应的齐次方程的全部特征值为 $\{(\frac{n\pi}{l})^2\}$ 和特征函数 $\{\sin \frac{n\pi x}{l}\}$, 将 u, f, φ, ψ 都按照特征函数系展开得

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}, \\ f(x, t) &= \sum_{n=1}^{\infty} f_n(t) \sin \frac{n\pi x}{l}, \\ \varphi(x) &= \sum_{n=1}^{\infty} \varphi_n \sin \frac{n\pi x}{l}, \\ \psi(x) &= \sum_{n=1}^{\infty} \psi_n \sin \frac{n\pi x}{l}. \end{aligned}$$

其中 f_n, φ_n, ψ_n 分别为 f, φ, ψ 的 Fourier 系数, $f_n = \frac{2}{l} \int_0^l f(x, t) \sin \frac{n\pi x}{l} dx$, 且 T_n 为 $u(x, t)$ 的 Fourier 系数, T_n 待定. 代入原方程得

$$\begin{cases} T_n'' + (\frac{n\pi}{l})^2 T_n = f_n(t), \\ T_n(0) = \varphi_n, \\ T_n'(0) = \psi_n. \end{cases}$$

解得

$$T_n(t) = \varphi_n \cos \frac{n\pi t}{l} + \frac{l}{n\pi} \psi_n \sin \frac{n\pi t}{l} + \frac{l}{n\pi} \int_0^t f_n(\tau) \sin \frac{n\pi(t-\tau)}{l} d\tau,$$

从而 $u(x, t)$ 具有形式解

$$u(x, t) = \sum_{n=1}^{\infty} (\varphi_n \cos \frac{n\pi t}{l} + \frac{l}{n\pi} \psi_n \sin \frac{n\pi t}{l} + \frac{l}{n\pi} \int_0^t f_n(\tau) \sin \frac{n\pi(t-\tau)}{l} d\tau) \sin \frac{n\pi x}{l}.$$

令 $F_n(t) := \frac{l}{n\pi} \int_0^t f_n(\tau) \sin \frac{n\pi(t-\tau)}{l} d\tau$, 而原问题齐次方程对应的 $u_n(x, t)$ 满足

$$\begin{aligned} |u_n(x, t)| &\leq o(\frac{1}{n^3}), \\ |Du_n(x, t)| &\leq o(\frac{1}{n^2}), \\ |D^2u_n(x, t)| &\leq a_n^2 + b_n^2 + O(\frac{1}{n^2}). \end{aligned}$$

因此由非齐次方程形式解的表达式只需考虑 $F_n \sin \frac{n\pi x}{l}$, $D(F_n \sin \frac{n\pi x}{l})$, $D^2(F_n \sin \frac{n\pi x}{l})$ 关于 n 的阶数.

假设 $f \in C^2[0, l]$, $f(0, t) = f(l, t) = 0$, 则通过分部积分得

$$f_n = \frac{2}{l} \int_0^l f(x, t) \sin \frac{n\pi x}{l} dx = O(\frac{C_n(t)}{n^2}), \text{ where } C_n(t) = \frac{2}{l} \int_0^l f_{xx}(x, t) \sin \frac{n\pi x}{l} dx.$$

从而 $F_n \sin \frac{n\pi x}{l} \leq o(\frac{1}{n^3})$, $D(F_n \sin \frac{n\pi x}{l}) \leq o(\frac{1}{n^2})$. 并且

$$\begin{aligned}
|D^2(F_n \sin \frac{n\pi x}{l})| &= |D^2(\frac{1}{n\pi} \int_0^t o(\frac{C_n}{n^2}) \sin \frac{n\pi(t-\tau)}{l} d\tau \sin \frac{n\pi x}{l})| \\
&= |D(\frac{1}{l} \int_0^t o(\frac{C_n}{n^2}) \sin \frac{n\pi(t-\tau)}{l} d\tau \cos \frac{n\pi x}{l})| \\
&= |\frac{\pi}{l^2} \int_0^t o(\frac{C_n}{n}) \sin \frac{n\pi(t-\tau)}{l} d\tau \sin \frac{n\pi x}{l}| \\
&\leq |o(\frac{TC_n}{n})|.
\end{aligned}$$

由 Bessel 不等式得

$$\sum_{n=1}^{\infty} C_n(t) \leq \frac{2}{l} \int_0^l f_{xx}^2(x, t) dx \leq M,$$

其中 $M < \infty$, 进一步有

$$\sum_{n=1}^N |\frac{TC_n}{n}| \leq \frac{T}{2} (\sum_{n=1}^N C_n^2)^{\frac{1}{2}} (\sum_{n=1}^N \frac{1}{n^2})^{\frac{1}{2}} \leq \frac{TM}{2} (\sum_{n=1}^N \frac{1}{n^2})^{\frac{1}{2}}.$$

当 $N \rightarrow \infty$ 时, $o(\frac{TC_n}{n})$ 收敛。此时非齐次方程得形式解为古典解。

□

四. PDE HOMEWORK 4

4.1 章二第 23 题

(4) 设 $u(x, t) = X(x)T(t) \neq 0$, 代入方程有

$$\begin{aligned} X(x)T''(t) - a^2 X''(x)T(t) &= 0, \\ \Leftrightarrow \frac{T''(t)}{a^2 T(t)} &= \frac{X''(x)}{X(x)} = -\lambda. \end{aligned}$$

从而

$$\begin{cases} T''(t) + a^2 \lambda T(t) = 0, \\ X''(x) + \lambda X(x) = 0. \end{cases}$$

由边界条件知

$$\begin{aligned} X'(0)T(t) &= (X'(l) + hX(l))T(t) = 0, \\ \Rightarrow X'(0) &= X'(l) + hX(l) = 0. \end{aligned}$$

得到如下 Sturm-Liouville 问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, & x \in (0, l), \\ X'(0) = 0, \\ X'(l) + hX(l) = 0. \end{cases}$$

Case 1: $h > 0$

运用定理 4.1, 此时 $\alpha_1 = 1, \beta_1 = 0, \alpha_2 = 1, \beta_2 = h$, 则 $\beta_1 + \beta_2 = h > 0$, 此时 $\lambda > 0$, 则解上述方程可得通解

$$X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x. \quad (25)$$

令 $\gamma := \sqrt{\lambda} > 0$, 则由 (25) 与边界条件可得

$$\begin{cases} C_1 \gamma = 0, \\ -C_2 \gamma \sin \gamma l + h C_2 \cos \gamma l = 0. \end{cases} \Rightarrow \begin{cases} C_1 = 0, \\ \tan \gamma l = \frac{h}{\gamma}. \end{cases}$$

记其正根为 $\gamma_1 < \gamma_2 < \dots$, 则特征值为 $\lambda_n = \gamma_n^2$, 故

$$X_n(x) = \cos \gamma_n x, \quad T_n(t) = A_n \sin a \gamma_n t + B_n \cos a \gamma_n t.$$

从而

$$u_n(x, t) = X_n(x)T_n(t) = (A_n \sin a \gamma_n t + B_n \cos a \gamma_n t) \cos \gamma_n x.$$

将所有 $u_n(x, t)$ 都加起来, 并使和满足初值条件. 记 $u(x, t) := \sum_{n=1}^{\infty} u_n(x, t)$, s.t.

$$\begin{cases} \varphi(x) = u|_{t=0} = \sum_{n=1}^{\infty} B_n \cos \gamma_n x, \\ \psi(x) = u_t|_{t=0} = \sum_{n=1}^{\infty} A_n a \gamma_n \cos \gamma_n x. \end{cases} \quad (26)$$

全体特征函数 $\{\cos\gamma_n x\}$ 组成了 $L^2[0, l]$ 上的完备正交系, 因此将 $\varphi(x), \psi(x)$ 按该正交系展开

$$\begin{cases} \varphi(x) = \sum_{n=1}^{\infty} \varphi_n \cos\gamma_n x, \text{ where } \varphi_n = \frac{\int_0^l \varphi(x) \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx}, \\ \psi(x) = \sum_{n=1}^{\infty} \psi_n \cos\gamma_n x, \text{ where } \psi_n = \frac{\int_0^l \psi(x) \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx}. \end{cases} \quad (27)$$

比较 (26) 与 (27) 得到

$$\begin{cases} A_n = \frac{1}{a\gamma_n} \psi_n = \frac{1}{a\gamma_n} \frac{\int_0^l \psi(x) \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx}, \\ B_n = \varphi_n = \frac{\int_0^l \varphi(x) \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx}. \end{cases}$$

综上所述得到

$$u(x, t) = \sum_{n=1}^{\infty} \left(\frac{1}{a\gamma_n} \frac{\int_0^l \psi(x) \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \sin a\gamma_n t + \frac{\int_0^l \varphi(x) \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \cos a\gamma_n t \right) \cos\gamma_n x.$$

(a) $\varphi(x) = \psi(x) = 1$

$$u(x, t) = \sum_{n=1}^{\infty} \left(\frac{1}{a\gamma_n} \frac{\int_0^l \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \sin a\gamma_n t + \frac{\int_0^l \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \cos a\gamma_n t \right) \cos\gamma_n x.$$

(b) $\varphi(x) = \cos \frac{\pi x}{l}, \psi(x) = 0$

$$u(x, t) = \sum_{n=1}^{\infty} \left(\frac{\int_0^l \cos \frac{\pi x}{l} \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \cos a\gamma_n t \right) \cos\gamma_n x.$$

Case 2: $h < 0$

当 $h < 0$ 时, 同理有如下 Sturm-Liouville 问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, & x \in (0, l), \\ X'(0) = 0, \\ X'(l) + hX(l) = 0, & h < 0. \end{cases}$$

对 λ 分以下三种情形讨论。

(i) $\lambda > 0$

Sturm-Liouville 问题的通解为

$$X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x.$$

令 $\gamma = \sqrt{\lambda}$, 再由边界条件可得

$$\begin{cases} C_1 \gamma = 0, \\ -C_2 \gamma \sin \gamma l + h C_2 \cos \gamma l = 0. \end{cases} \Rightarrow \begin{cases} C_1 = 0, \\ \tan \gamma l = \frac{h}{\gamma}, \quad h < 0. \end{cases}$$

记其正根为 $\gamma_1 < \gamma_2 < \dots$, 则特征值为 $\lambda_n = \gamma_n^2$, 故

$$X_n(x) = \cos \gamma_n x, \quad T_n(t) = A_n \sin a\gamma_n t + B_n \cos a\gamma_n t.$$

从而

$$u_n(x, t) = X_n(x) T_n(t) = (A_n \sin a\gamma_n t + B_n \cos a\gamma_n t) \cos \gamma_n x.$$

将所有 $u_n(x, t)$ 都加起来, 并使和满足初值条件。记 $u(x, t) := \sum_{n=1}^{\infty} u_n(x, t)$, s.t.

$$\begin{cases} \varphi(x) = u|_{t=0} = \sum_{n=1}^{\infty} B_n \cos \gamma_n x, \\ \psi(x) = u_t|_{t=0} = \sum_{n=1}^{\infty} A_n a \gamma_n \cos \gamma_n x. \end{cases} \quad (28)$$

全体特征函数 $\{\cos \gamma_n x\}$ 组成了 $L^2[0, l]$ 上的完备正交系, 因此将 $\varphi(x)$, $\psi(x)$ 按该正交系展开

$$\begin{cases} \varphi(x) = \sum_{n=1}^{\infty} \varphi_n \cos \gamma_n x, \text{ where } \varphi_n = \frac{\int_0^l \varphi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx}, \\ \psi(x) = \sum_{n=1}^{\infty} \psi_n \cos \gamma_n x, \text{ where } \psi_n = \frac{\int_0^l \psi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx}. \end{cases} \quad (29)$$

比较 (28) 与 (29) 得到

$$\begin{cases} A_n = \frac{1}{a \gamma_n} \psi_n = \frac{1}{a \gamma_n} \frac{\int_0^l \psi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx}, \\ B_n = \varphi_n = \frac{\int_0^l \varphi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx}. \end{cases}$$

从而

$$T(t) = \frac{1}{a \gamma_n} \frac{\int_0^l \psi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx} \sin a \gamma_n t + \frac{\int_0^l \varphi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx} \cos a \gamma_n t.$$

(ii) $\lambda < 0$

同理代入 Sturm-Liouville 问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, \quad x \in (0, l), \\ X'(0) = 0, \\ X'(l) + hX(l) = 0. \end{cases}$$

解得 $X(x) = C_1 e^{\sqrt{-\lambda}x} + C_2 e^{-\sqrt{-\lambda}x}$, 代入边界条件得

$$\begin{cases} C_1 \sqrt{-\lambda} - C_2 \sqrt{-\lambda} = 0, \\ (h + \sqrt{-\lambda})C_1 e^{\sqrt{-\lambda}l} + (h - \sqrt{-\lambda})C_2 e^{-\sqrt{-\lambda}l} = 0. \end{cases} \Rightarrow \begin{cases} C_1 = 0, \\ C_2 = 0. \end{cases}$$

从而 $X(x) = 0$, 矛盾!

(iii) $\lambda = 0$

同理代入 Sturm-Liouville 问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, \quad x \in (0, l), \\ X'(0) = 0, \\ X'(l) + hX(l) = 0. \end{cases}$$

解得 $X(x) = 0$, 矛盾!

综上, 当 $h < 0$ 时, 原定解问题的解为

$$u(x, t) = \sum_{n=1}^{\infty} \left(\frac{1}{a \gamma_n} \frac{\int_0^l \psi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx} \sin a \gamma_n t + \frac{\int_0^l \varphi(x) \cos \gamma_n x dx}{\int_0^l \cos^2 \gamma_n x dx} \cos a \gamma_n t \right) \cos \gamma_n x.$$

(a) $\varphi(x) = \psi(x) = 1$

$$u(x, t) = \sum_{n=1}^{\infty} \left(\frac{1}{a\gamma_n} \frac{\int_0^l \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \sin a\gamma_n t + \frac{\int_0^l \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \cos a\gamma_n t \right) \cos\gamma_n x.$$

(b) $\varphi(x) = \cos \frac{\pi x}{l}, \psi(x) = 0$

$$u(x, t) = \sum_{n=1}^{\infty} \left(\frac{\int_0^l \cos \frac{\pi x}{l} \cos\gamma_n x dx}{\int_0^l \cos^2\gamma_n x dx} \cos a\gamma_n t \right) \cos\gamma_n x.$$

Case 3: $h=0$

当 $\lambda = 0$ 时, $X(x)$ 为常数, 不妨设为 $X(x) = 1$ 。当 $\lambda > 0$ 时, 通解 $X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x$, 由边界条件可得

$$\begin{cases} C_1 \sqrt{\lambda} = 0, \\ C_2 \sqrt{\lambda} \sin \sqrt{\lambda} l = 0. \end{cases}$$

解得 $C_1 = 0, \lambda_n = (\frac{n\pi}{l})^2, n \in \mathbb{N}_+$, 从而 $X(x) = C \cos \frac{n\pi}{l} x, n \in \mathbb{N}$, 进一步有

$$T_n''(t) + (\frac{an\pi}{l})^2 T_n(t) = 0.$$

解得

$$T_n(t) = \begin{cases} A_n \sin \frac{an\pi}{l} t + B_n \cos \frac{an\pi}{l} t, & n > 0, \\ at + b, & n = 0. \end{cases}$$

故

$$u_n(x, t) = X_n(x) T_n(t) = \begin{cases} (A_n \sin \frac{an\pi}{l} t + B_n \cos \frac{an\pi}{l} t) \cos \frac{n\pi}{l} x, & n > 0, \\ at + b, & n = 0. \end{cases}$$

将所有 $u_n(x, t)$ 都加起来, 并使和满足初值条件。记 $u(x, t) := \sum_{n=1}^{\infty} u_n(x, t), s.t.$

$$\begin{cases} \varphi(x) = u|_{t=0} = b + \sum_{n=1}^{\infty} B_n \cos \frac{n\pi}{l} x, \\ \psi(x) = u_t|_{t=0} = a + \sum_{n=1}^{\infty} A_n \frac{an\pi}{l} \cos \frac{n\pi}{l} x. \end{cases} \quad (30)$$

全体特征函数 $\{\cos \frac{n\pi}{l} x\}$ 组成了 $L^2[0, l]$ 上的完备正交系, 因此将 $\varphi(x), \psi(x)$ 按该正交系展开

$$\begin{cases} \varphi(x) = \sum_{n=1}^{\infty} \varphi_n \cos \frac{n\pi}{l} x, \text{ where } \varphi_n = \frac{2}{l} \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx, \varphi_0 = \frac{1}{l} \int_0^l \varphi(x) dx, \\ \psi(x) = \sum_{n=1}^{\infty} \psi_n \cos \frac{n\pi}{l} x, \text{ where } \psi_n = \frac{2}{l} \int_0^l \psi(x) \cos \frac{n\pi}{l} x dx, \psi_0 = \frac{1}{l} \int_0^l \psi(x) dx. \end{cases} \quad (31)$$

比较 (30) 与 (31) 得到

$$\begin{cases} A_n = \frac{l}{an\pi} \psi_n = \frac{l}{an\pi} \frac{2}{l} \int_0^l \psi(x) \cos \frac{n\pi}{l} x dx, a = \frac{1}{l} \int_0^l \psi(x) dx, \\ B_n = \frac{2}{l} \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx, b = \frac{1}{l} \int_0^l \varphi(x) dx. \end{cases}$$

综上所述得到

$$u(x, t) = \frac{t}{l} \int_0^l \psi(x) dx + \frac{1}{l} \int_0^l \varphi(x) dx + \frac{2}{l} \sum_{n=1}^{\infty} \left(\frac{l}{an\pi} \int_0^l \psi(x) \cos \frac{n\pi}{l} x dx \sin \frac{an\pi}{l} t + \int_0^l \varphi(x) \cos \frac{n\pi}{l} x dx \cos \frac{an\pi}{l} t \right) \cos \frac{n\pi}{l} x.$$

(a) $\varphi(x) = \psi(x) = 1$ 当 $n \geq 1$ 时有 $\int_0^l \cos \frac{n\pi}{l} x dx = \frac{1}{n\pi} \int_0^{n\pi} \cos x dx = \frac{\sin n\pi}{n\pi} = 0$, 故

$$u(x, t) = t + 1.$$

(b) $\varphi(x) = \cos \frac{\pi x}{l}$, $\psi(x) = 0$

$$\begin{aligned} u(x, t) &= \frac{1}{l} \int_0^l \cos \frac{\pi x}{l} x dx + \sum_{n=1}^{\infty} \left(\frac{2}{l} \int_0^l \cos \frac{\pi x}{l} \cos \frac{n\pi}{l} x dx \cos \frac{an\pi}{l} t \right) \cos \frac{n\pi}{l} x \\ &= \frac{2}{l} \left(\int_0^l \left(\cos \frac{\pi x}{l} \right)^2 dx \cos \frac{a\pi}{l} t \right) \cos \frac{\pi}{l} x \\ &= \frac{2}{l} \left(\frac{l}{2} \cos \frac{a\pi}{l} t \right) \cos \frac{\pi}{l} x \\ &= \cos \frac{a\pi}{l} t \cos \frac{\pi}{l} x. \end{aligned}$$

□

4.2 章二第 28 题

考虑如下方程组

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), & (x, t) \in (0, l) \times (0, \infty), \\ u|_{t=0} = \varphi(x), & u_t|_{t=0} = \psi(x), \quad x \in [0, l], \\ -u_x + \alpha u|_{x=0} = 0, & t \geq 0. \\ u_x + \beta u|_{x=l} = 0, & t \geq 0. \end{cases} \quad (32)$$

记 $K_\tau := \{(x, t) | 0 \leq t \leq \tau, 0 \leq x \leq l\}$, $\Omega_\tau := \{(x, t) | t = \tau, 0 \leq x \leq l\}$, 并设 u_1, u_2 为 (32) 的解, 则 $\tilde{u} := u_1 - u_2$ 满足方程组

$$\begin{cases} \tilde{u}_{tt} - a^2 \tilde{u}_{xx} = 0, & (x, t) \in (0, l) \times (0, \infty), \\ \tilde{u}|_{t=0} = 0, \quad \tilde{u}_t|_{t=0} = 0, & x \in [0, l], \\ -\tilde{u}_x + \alpha \tilde{u}|_{x=0} = 0, & t \geq 0, \\ \tilde{u}_x + \beta \tilde{u}|_{x=l} = 0, & t \geq 0. \end{cases} \quad (33)$$

在 (33) 波动方程两边同乘 $\tilde{u}_t$ 并在 K_τ 上积分得

$$\begin{aligned} J &:= \iint_{K_\tau} (\tilde{u}_{tt} - a^2 \tilde{u}_{xx}) \tilde{u}_t dx dt = 0 \\ &= - \oint_{\partial K_\tau} (a^2 \tilde{u}_t \tilde{u}_x dt + \frac{1}{2} (\tilde{u}_t^2 + a^2 \tilde{u}_x^2) dx) \\ \Rightarrow -J &= \left(\int_{\Omega_\tau} + \int_{\Omega_0} + \int_{\Gamma_{\tau 1} \cup \Gamma_{\tau 2}} \right) (a^2 \tilde{u}_t \tilde{u}_x dt + \frac{1}{2} (\tilde{u}_t^2 + a^2 \tilde{u}_x^2) dx) \\ &:= J_1 + J_2 + J_3. \end{aligned}$$

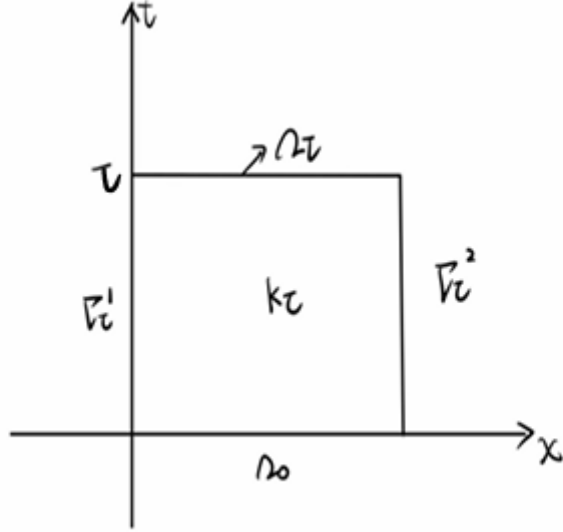


图 4: pic5

其中 Ω_0 , $\Gamma_{\tau 1}$, $\Gamma_{\tau 2}$ 如图所示

注意到

$$\begin{aligned}
 J_1 &= \int_0^l \frac{1}{2} (\tilde{u}_t^2(x, \tau) + a^2 \tilde{u}_x^2(x, \tau)) dx, \\
 J_2 &= - \int_0^l \frac{1}{2} (\tilde{u}_t^2(x, 0) + a^2 \tilde{u}_x^2(x, 0)) dx \\
 &= - \int_0^l \frac{1}{2} (\psi_t^2(x, 0) + a^2 \varphi_x^2(x, 0)) dx \\
 &= 0, \\
 J_3 &= -a^2 \int_0^\tau (-\beta \tilde{u}(l, t) \tilde{u}_t(l, t) - \alpha \tilde{u}(0, t) \tilde{u}_t(0, t)) dt \\
 &= \frac{a^2}{2} \beta (\tilde{u}^2(l, \tau) - \tilde{u}^2(l, 0)) + \frac{a^2}{2} \alpha (\tilde{u}^2(0, \tau) - \tilde{u}^2(0, 0)) \\
 &\geq 0.
 \end{aligned}$$

因此

$$\begin{aligned}
 0 = J &= J_1 + J_3 \geq J_1 = \int_0^l \frac{1}{2} (\tilde{u}_t^2(x, \tau) + a^2 \tilde{u}_x^2(x, \tau)) dx \geq 0, \\
 \Rightarrow \quad \tilde{u}_t(x, \tau) &= \tilde{u}_x(x, \tau) = 0, \quad \forall \tau \geq 0.
 \end{aligned}$$

再对 $\tilde{u}$ 进行估计,

$$\begin{aligned}
 \iint_{K_\tau} \tilde{u} \tilde{u}_t dx dt &= \int_0^\tau \int_0^l \tilde{u} \tilde{u}_t dx dt \\
 &= \frac{1}{2} \int_0^\tau \int_0^l (\tilde{u})_t^2 dx dt \\
 &= \frac{1}{2} \int_0^l \tilde{u}^2(x, \tau) dx.
 \end{aligned}$$

又由 Young 不等式知 $\int_0^\tau \int_0^l \tilde{u} \tilde{u}_t dx dt \leq \frac{1}{2} \int_0^\tau \int_0^l \tilde{u}^2 + \tilde{u}_t^2 dx dt$, 则

$$\frac{1}{2} \int_0^l \tilde{u}^2(x, \tau) dx \leq \frac{1}{2} \int_0^\tau \int_0^l \tilde{u}^2 + \tilde{u}_t^2 dx dt.$$

令 $G(\tau) := \int_0^\tau \int_0^l \tilde{u}^2 dx dt$, $F(\tau) := \int_0^\tau \int_0^l \tilde{u}_t^2 dx dt = 0$, 则

$$\frac{dG(\tau)}{d\tau} = \int_0^l \tilde{u}^2(x, \tau) dx \leq G(\tau) + F(\tau),$$

由 Gronwall 不等式得

$$G(\tau) \leq C(\tau)F(\tau) = 0,$$

$$\Rightarrow \quad \tilde{u} = 0.$$

唯一性得证。

□

4.3 章三第 1 题

(1)

$$\begin{aligned} \hat{f}(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\lambda x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a |x| e^{-i\lambda x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a |x| (\cos \lambda x - i \sin \lambda x) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a |x| \cos \lambda x dx \\ &= \frac{2}{\sqrt{2\pi}} \int_0^a x \cos \lambda x dx \\ &= \frac{2}{\sqrt{2\pi}} \frac{1}{\lambda} (x \sin \lambda x \Big|_0^a - \int_0^a \sin \lambda x dx) \\ &= \frac{2}{\sqrt{2\pi}} \left(\frac{a}{\lambda} \sin \lambda a + \frac{1}{\lambda^2} (\cos \lambda a - 1) \right). \end{aligned}$$

□

4.4 章三第 2 题

(8) 令 $g_1(x) := e^{-a|x|}$, 同理 p115 例 3 可得

$$\begin{aligned} \hat{g}_1(\lambda) &= \frac{1}{\sqrt{2\pi}} \frac{2}{a^2 + \lambda^2}, \\ \Rightarrow \quad \left(\frac{1}{a^2 + \lambda^2} \right)^\vee &= \frac{1}{a} \frac{\sqrt{2\pi}}{2} e^{-a|x|}. \end{aligned}$$

令 $g_2(x) := \frac{1}{a^2 + x^2}$, 则

$$\begin{aligned}\hat{g}_2(\lambda) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{a^2 + x^2} e^{-i\lambda x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{a^2 + x^2} e^{i\lambda x} dx \\ &= \left(\frac{1}{a^2 + x^2} \right)^{\vee} \\ &= \frac{1}{a} \frac{\sqrt{2\pi}}{2} e^{-a|\lambda|}.\end{aligned}$$

再由乘多项式的性质得

$$\hat{f}(\lambda) = (xg_2(x))^{\wedge} = i \frac{d}{d\lambda} \hat{g}_2(\lambda) = -\frac{\sqrt{2\pi}}{2} i e^{-a|\lambda|} \text{sign}(\lambda).$$

□

4.5 章三第 3 题

(3)

$$\begin{aligned}\check{f}(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-|\lambda|y} e^{i\lambda x} d\lambda \\ &= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^0 e^{\lambda(ix+y)} d\lambda + \int_0^{\infty} e^{-\lambda(-ix+y)} d\lambda \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{ix+y} e^{\lambda(ix+y)} \Big|_{-\infty}^0 + \frac{1}{ix-y} e^{-\lambda(-ix+y)} \Big|_0^{\infty} \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{ix+y} + \frac{1}{-ix+y} \right) \\ &= \frac{2}{\sqrt{2\pi}} \frac{y}{x^2 + y^2}.\end{aligned}$$

□

4.6 章三第 4 题

(1) 对方程两边关于 x 做 Fourier 变换得

$$\begin{cases} \frac{\partial \hat{u}}{\partial t} + a^2 \lambda^2 \hat{u} - b\lambda i \hat{u} - c\hat{u} = \hat{f}(\lambda, t), & -\infty \leq \lambda \leq \infty, t > 0, \\ \hat{u}|_{t=0} = \hat{\varphi}(\lambda), & -\infty \leq \lambda \leq \infty. \end{cases}$$

解得

$$\hat{u}(\lambda, t) = \hat{\varphi} e^{(-a^2 \lambda^2 + ib\lambda + c)t} + \int_0^t \hat{f}(\lambda, \tau) e^{(-a^2 \lambda^2 + ib\lambda + c)(t-\tau)} d\tau, \quad (34)$$

对 (34) 两边做反演得

$$\hat{u}(\lambda, t)^{\vee} = (\hat{\varphi} e^{(-a^2 \lambda^2 + ib\lambda + c)t})^{\vee} + \int_0^t (\hat{f}(\lambda, \tau) e^{(-a^2 \lambda^2 + ib\lambda + c)(t-\tau)})^{\vee} d\tau.$$

下求 $g(t)$ 使得 $\hat{g}(\lambda) = e^{(-a^2 \lambda^2 + ib\lambda + c)t}$ 。计算得到

$$g(t) = \frac{1}{\sqrt{2a^2 t}} e^{ct - \frac{(x+bt)^2}{4a^2 t}}.$$

由卷积性质

$$\begin{aligned} ((f * g)(t))^{\wedge} &= \sqrt{2\pi} \hat{f} \hat{g}, \\ \Rightarrow (\hat{f} \hat{g})^{\vee} &= \frac{1}{\sqrt{2\pi}} (f * g)(t). \end{aligned}$$

得到

$$\begin{aligned} u(x, t) &= \frac{1}{\sqrt{2\pi}} (\varphi * g + \int_0^t f * g(x, t - \tau) d\tau) \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\sqrt{2a^2 t}} \int_{-\infty}^{\infty} \varphi(\xi) e^{ct - \frac{(x - \xi + bt)^2}{4a^2 t}} d\xi + \int_0^t \frac{1}{\sqrt{2a^2(t - \tau)}} \int_{-\infty}^{\infty} f(\xi, \tau) e^{c(t - \tau) - \frac{(x - \xi + b(t - \tau))^2}{4a^2(t - \tau)}} d\xi d\tau \right). \end{aligned}$$

□

五. PDE HOMEWORK 5

5.1 章三第 5 题

(2) 对 $\forall \psi \in \mathcal{D}$, 有

$$\begin{aligned} \langle \varphi(x)\delta'(x), \psi \rangle &= \langle \delta'(x), \varphi(x)\psi(x) \rangle \\ &= - \langle \delta(x), (\varphi\psi)' \rangle \\ &= - \langle \delta(x), \varphi\psi' + \varphi'\psi \rangle \\ &= -\varphi(0)\psi'(0) + \varphi'(0)\psi(0) \\ &= \varphi(0) \langle \delta'(x), \psi(x) \rangle - \varphi'(0) \langle \delta(x), \psi(x) \rangle \\ &= \langle \varphi(0)\delta'(x) - \varphi'(0)\delta(x), \psi(x) \rangle. \end{aligned}$$

因此 $\varphi(x)\delta'(x) = \varphi(0)\delta'(x) - \varphi'(0)\delta(x)$ 。

□

5.2 章三第 7 题

(3) 对 $\forall \varphi \in \mathcal{D}(\mathbb{R})$, 有

$$\begin{aligned} \langle f', \varphi \rangle &= - \langle f, \varphi' \rangle \\ &= - \int_{-\infty}^{\infty} f(x)\varphi'(x)dx \\ &= - \int_{-1}^1 x^2\varphi'(x)dx \\ &= -(x^2\varphi(x))|_{-1}^1 - \int_{-1}^1 2x\varphi(x)dx \\ &= -\varphi(1) + \varphi(-1) + \int_{-\infty}^{\infty} 2x(H(x+1) - H(x-1))\varphi(x)dx \\ &= \langle -\delta(x-1) + \delta(x+1) + 2x(H(x+1) - H(x-1)), \varphi(x) \rangle. \end{aligned}$$

其中 $H(x)$ 为 Heaviside 函数。故

$$f'(x) = -\delta(x-1) + \delta(x+1) + 2x(H(x+1) - H(x-1)).$$

□

5.3 章三第 9 题

(5) 令 $v = u - x$, 则原方程化为

$$\begin{cases} v_t - a^2 v_{xx} = x(l-x) := f(x), & 0 < x < l, t > 0, \\ v|_{t=0} = \sin \frac{\pi x}{l} - x := \varphi(x), & 0 \leq x \leq l, \\ v|_{x=0} = v_x|_{x=l} = 0, & t > 0. \end{cases} \quad (35)$$

考虑形如 $v(x, t) = X(x)T(t)$ 的解, 代入 (35) 得 Sturm-Liouville 问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l, \\ X(0) = X'(l) = 0. \end{cases}$$

其中 $X''(x) + \lambda X(x) = 0$ 的通解为

$$X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x,$$

代入边界条件得

$$C_2 = 0, \quad C_1 \sqrt{\lambda} \cos \sqrt{\lambda} l = 0.$$

因此, $\sqrt{\lambda_n} = \frac{(n + \frac{1}{2})\pi}{l}$, $X_n(x) = \sin \frac{(n + \frac{1}{2})\pi x}{l}$, 从而

$$\begin{cases} v(x, t) = \sum_{n=0}^{\infty} T_n(t) \sin \frac{(n + \frac{1}{2})\pi x}{l}, \\ f(x) = \sum_{n=0}^{\infty} f_n \sin \frac{(n + \frac{1}{2})\pi x}{l}, \quad f_n = \frac{2}{l} \int_0^l f(x) \sin \frac{(n + \frac{1}{2})\pi x}{l} dx, \\ \varphi(x) = \sum_{n=0}^{\infty} \varphi_n \sin \frac{(n + \frac{1}{2})\pi x}{l}, \quad \varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{(n + \frac{1}{2})\pi x}{l} dx. \end{cases} \quad (36)$$

将 (36) 代入原方程得

$$\begin{cases} T'_n + \left(\frac{a(n + \frac{1}{2})\pi}{l}\right)^2 T_n = f_n, \\ T_n(0) = \varphi_n. \end{cases}$$

解得

$$T_n(t) = \varphi_n e^{-\left(\frac{a(n + \frac{1}{2})\pi}{l}\right)^2 t} + \int_0^t f_n e^{-\left(\frac{a(n + \frac{1}{2})\pi}{l}\right)^2 (t - \tau)} d\tau,$$

故

$$\begin{aligned} v(x, t) &= \sum_{n=0}^{\infty} v_n(x, t) \\ &= \sum_{n=0}^{\infty} \left(\varphi_n e^{-\left(\frac{a(n + \frac{1}{2})\pi}{l}\right)^2 t} + \int_0^t f_n e^{-\left(\frac{a(n + \frac{1}{2})\pi}{l}\right)^2 (t - \tau)} d\tau \right) \sin \frac{(n + \frac{1}{2})\pi x}{l}, \end{aligned}$$

其中 f_n, φ_n 的定义同 (36)。

□

5.4 章三第 10 题

(1) 令 $v := u - w = u - \frac{h(U_2 - U_1)x}{2 + hl} - \frac{U_2 - U_1}{2 + hl} - U_1$, 则 v 满足

$$\begin{cases} v_t - a^2 v_{xx} = 0, & 0 < x < l, t > 0, \\ v|_{t=0} = \varphi(x) - w, & 0 \leq x \leq l, \\ (-v_x + hv)|_{x=0} = 0, & t > 0, \\ (v_x + hv)|_{x=l} = 0, & t > 0. \end{cases} \quad (37)$$

考虑形为 $v(x, t) = X(x)T(t)$ 的解, 代入 (37) 得 Sturm-Liouville 问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l, \\ -X'(0) + hX(0) = 0, \\ X'(l) + hX(l) = 0. \end{cases}$$

其中 $X''(x) + \lambda X(x) = 0$ 的通解为

$$X(x) = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x,$$

代入边界条件得

$$\begin{cases} -C_1 \sqrt{\lambda} + hC_2 = 0, \\ (C_1 \sqrt{\lambda} + hC_2) \cos \sqrt{\lambda} l + (hC_1 - C_2 \sqrt{\lambda}) \sin \sqrt{\lambda} l = 0. \end{cases}$$

令 $\sqrt{\lambda} = \beta$, $C_1 = h$, 则 $C_2 = \beta$, $\tan \beta l = \frac{2h\beta}{\beta^2 - h^2}$. 所有特征值组成了单调递增的序列, 不妨记为 $0 \leq \beta_1^2 < \beta_2^2 < \dots < \beta_n^2 < \dots$, 与特征值对应的特征函数为

$$X_n(x) = h \sin \beta_n x + \beta_n \cos \beta_n x.$$

将 $X_n(x)$ 代入 (37) 得

$$\begin{cases} T'_n + (a\beta_n)^2 T_n = 0, \\ T_n(0) = C_n. \end{cases}$$

其中 C_n 由初值条件 $v(x, t)|_{t=0} = \varphi(x) - w(0)$ 确定

$$\Rightarrow C_n = \frac{\int_0^l (h \sin \beta_n x + \beta_n \cos \beta_n x) (\varphi(x) - w(0)) dx}{\int_0^l (h \sin \beta_n x + \beta_n \cos \beta_n x)^2 dx}.$$

解得

$$T_n = C_n e^{-(a\beta_n)^2 t}.$$

因此

$$v(x, t) = \sum_{n=0}^{\infty} C_n e^{-(a\beta_n)^2 t} (h \sin \beta_n x + \beta_n \cos \beta_n x). \quad (38)$$

当 (38) 一致收敛时,

$$\lim_{t \rightarrow \infty} v(x, t) = \sum_{n=0}^{\infty} \lim_{t \rightarrow \infty} C_n e^{-(a\beta_n)^2 t} (h \sin \beta_n x + \beta_n \cos \beta_n x) = 0,$$

综上,

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} v(x, t) - w(x) = -w(x) = \frac{h(U_2 - U_1)x}{2 + hl} + \frac{U_2 - U_1}{2 + hl} + U_1.$$

□

5.5 章三第 11 题

方法一：在 $0 < x < l_1 + l_2$ 上拼接两杆。

由题意知，设组合杆上的温度分布为 $u(x, t)$ 满足以下混合问题

$$\begin{cases} u_t - a_1^2 u_{xx} = 0, & 0 < x < l_1, t > 0, \\ u_t - a_2^2 u_{xx} = 0, & l_1 < x < l_1 + l_2, t > 0, \\ u|_{t=0} = \varphi(x), & 0 \leq x \leq l_1 + l_2, \\ u|_{x=0} = u|_{x=l_1+l_2} = 0, & t > 0, \\ u(l_1 - 0, t) = u(l_1 + 0, t), & t > 0, \\ k_1 u_x(l_1 - 0, t) = k_2 u_x(l_1 + 0, t), & t > 0. \end{cases} \quad (39)$$

其中 $a_i^2 = \frac{k_i}{c_i \rho}$, $i = 1, 2$, 且 $u(l_1 - 0, t)$, $u(l_1 + 0, t)$ 分别代表左右极限。设 $u(l_1 - 0, t) = u(l_1 + 0, t) = g(t)$, 则 (39) 中 $0 < x < l_1$ 部分化为

$$\begin{cases} u_t - a_1^2 u_{xx} = 0, & 0 < x < l_1, t > 0, \\ u|_{t=0} = \varphi(x), & 0 \leq x \leq l_1, \\ u|_{x=0} = 0, & t > 0, \\ u|_{x=l_1} = g(t), & t > 0. \end{cases} \quad (40)$$

令 $v = u - \frac{x}{l_1} g(t)$, 代入 (40) 得

$$\begin{cases} v_t - a_1^2 v_{xx} = -\frac{x}{l_1} g'(t) := f(x, t), & 0 < x < l_1, t > 0, \\ v|_{t=0} = \varphi(x) - \frac{x}{l_1} g(0) := \psi(x), & 0 \leq x \leq l_1, \\ v|_{x=0} = v|_{x=l_1} = 0, & t > 0. \end{cases}$$

由分离变量法解得

$$v(x, t) = \sum_{n=0}^{\infty} \left(\psi_n e^{-\left(\frac{n\pi a_1}{l_1}\right)^2 t} + \int_0^t f_n(\tau) e^{-\left(\frac{n\pi a_1}{l_1}\right)^2 (t-\tau)} d\tau \right) \sin \frac{n\pi x}{l_1},$$

其中

$$\begin{aligned} f_n &= \frac{2}{l_1} \int_0^{l_1} f(x, t) \sin \frac{n\pi x}{l_1} dx, \quad \psi_n = \frac{2}{l_1} \int_0^{l_1} \psi(x) \sin \frac{n\pi x}{l_1} dx, \\ \Rightarrow \quad u(x, t) &= v(x, t) + \frac{x}{l_1} g(t), \quad 0 < x < l_1. \end{aligned} \quad (41)$$

进一步地, (39) 中杆在 $l_1 < x < l_1 + l_2$ 部分可视作在 $0 < x < l_2$ 上的杆向右平移 l_1 得到。因此考虑 $0 < x < l_2$ 部分

$$\begin{cases} u_t - a_2^2 u_{xx} = 0, & 0 < x < l_2, t > 0, \\ u|_{t=0} = \varphi(x), & 0 \leq x \leq l_2, \\ u|_{x=0} = g(t), & t > 0, \\ u|_{x=l_2} = 0, & t > 0. \end{cases} \quad (42)$$

同理令 $w = u - \frac{l_2 - x}{l_2}g(t)$, 代入 (42) 得

$$\begin{cases} w_t - a_2^2 w_{xx} = -\frac{l_2 - x}{l_2}g'(t) := \tilde{f}(x, t), & 0 < x < l_2, t > 0, \\ w|_{t=0} = \varphi(x) - \frac{l_2 - x}{l_2}g(0) := \tilde{\psi}(x), & 0 \leq x \leq l_2, \\ w|_{x=0} = w|_{x=l_2} = 0, & t > 0. \end{cases}$$

同理由分离变量法解得

$$w(x, t) = \sum_{n=0}^{\infty} \left(\tilde{\psi}_n e^{-\left(\frac{n\pi a_2}{l_2}\right)^2 t} + \int_0^t \tilde{f}_n(\tau) e^{-\left(\frac{n\pi a_2}{l_2}\right)^2 (t-\tau)} d\tau \right) \sin \frac{n\pi x}{l_2},$$

其中

$$\begin{aligned} \tilde{f}_n &= \frac{2}{l_2} \int_0^{l_2} \tilde{f}(x, t) \sin \frac{n\pi x}{l_2} dx, \quad \tilde{\psi}_n = \frac{2}{l_2} \int_0^{l_2} \tilde{\psi}(x) \sin \frac{n\pi x}{l_2} dx, \\ \Rightarrow \quad u(x, t) &= w(x - l_1, t) + \frac{l_1 + l_2 - x}{l_2} g(t), \quad l_1 < x < l_1 + l_2. \end{aligned} \quad (43)$$

对 (41) 和 (43), 由连接条件知

$$\begin{aligned} \left(v(x, t) + \frac{x}{l_1} g(t) \right) \Big|_{x=l_1} &= \left(w(x - l_1, t) + \frac{l_1 + l_2 - x}{l_2} g(t) \right) \Big|_{x=l_1}, \\ \Leftrightarrow \quad v(l_1, t) &= w(0, t). \end{aligned}$$

此外由 (39) 的初速度条件知还需要满足

$$\begin{aligned} k_1 \left(v(x, t) + \frac{x}{l_1} g(t) \right) \Big|_{x=l_1} &= k_2 \left(w(x - l_1, t) + \frac{l_1 + l_2 - x}{l_2} g(t) \right) \Big|_{x=l_1}, \\ \Rightarrow \quad k_1 \left(v_x(l_1, t) + \frac{1}{l_1} g(t) \right) &= k_2 \left(w_x(0, t) - \frac{1}{l_2} g(t) \right), \end{aligned}$$

$$\begin{aligned} \Rightarrow g(t) &= \frac{l_1 l_2}{k_1 l_2 + k_2 l_1} (k_2 w_x(0, t) - k_1 v_x(l_1, t)), \\ &= \frac{l_1 l_2}{k_1 l_2 + k_2 l_1} \sum_{n=0}^{\infty} \left(\frac{n k_2 \pi}{l_2} \left(\tilde{\psi}_n e^{-\left(\frac{n\pi a_2}{l_2}\right)^2 t} + \int_0^t \tilde{f}_n(\tau) e^{-\left(\frac{n\pi a_2}{l_2}\right)^2 (t-\tau)} d\tau \right) \right. \\ &\quad \left. - \frac{n \pi k_1}{l_1} \cos n \pi \left(\psi_n e^{-\left(\frac{n\pi a_1}{l_1}\right)^2 t} + \int_0^t f_n(\tau) e^{-\left(\frac{n\pi a_1}{l_1}\right)^2 (t-\tau)} d\tau \right) \right). \end{aligned}$$

综上,

$$u(x, t) = \begin{cases} v(x, t) + \frac{x}{l_1} g(t), & 0 < x < l_1, t > 0 \\ w(x - l_1, t) + \frac{l_1 + l_2 - x}{l_2} g(t), & l_1 < x < l_1 + l_2, t > 0. \end{cases}$$

方法二: 在 $-l_1 < x < l_2$ 上拼接两杆。

做法同上。

□

5.6 章三第 18 题

假设 $(x^*, y^*) \in \overline{Q}$, s.t. $u(x^*, y^*) = \max_{\overline{Q}} u(x, t) \geq 0$, 分两种情况讨论.

(1) 当 $Lu < 0$ 时, 反设 $(x^*, y^*) \notin \Gamma$, i.e. $(x^*, y^*) \in Q$, 根据极值的必要条件可得

$$\begin{cases} u_t|_{(x^*, t^*)} = 0, \\ u_x|_{(x^*, t^*)} = 0, \\ u_{xx}|_{(x^*, t^*)} \leq 0, \end{cases}$$

因此

$$Lu|_{(x^*, t^*)} = (u_t - a^2 u_{xx} + c(x, t)u)|_{(x^*, t^*)} \geq 0$$

与 $Lu < 0$ 矛盾!

(2) 当 $Lu \leq 0$ 时, 引入辅助函数 $v = u + \epsilon e^{-C_0 t}$, ($\epsilon > 0$) 且 $|c(x, t)| \leq M$, 则

$$\begin{aligned} Lv &= L(u + \epsilon e^{-C_0 t}) \\ &= Lu - C_0 \epsilon e^{-C_0 t} + c(x, t) \epsilon e^{-C_0 t}, \end{aligned}$$

令其中的 $C_0 = 2M$, 则

$$Lv = L_u + (c(x, t) - 2M) \epsilon e^{-2Mt} < 0.$$

由 (1) 知 $\max_{\overline{Q}} v(x, t) = \max_{\Gamma} v(x, t)$, 进一步地

$$\max_{\overline{Q}} u = \max_{\overline{Q}} (v - \epsilon e^{-C_0 t}) \leq \max_{\overline{Q}} v - \epsilon e^{-C_0 T} \leq \max_{\Gamma} v - \epsilon e^{-C_0 T},$$

即 $\max_{\overline{Q}} u + \epsilon e^{-C_0 T} \leq \max_{\Gamma} v$, 又因为 $\max_{\Gamma} u \leq \max_{\overline{Q}} u$, 则

$$\max_{\Gamma} u \leq \max_{\overline{Q}} u \leq \max_{\overline{Q}} u + \epsilon e^{-C_0 T} \leq \max_{\Gamma} v.$$

令 $\epsilon \rightarrow 0$, 又因为 $e^{-C_0 t}$ 有界, 则

$$\max_{\Gamma} u \leq \max_{\overline{Q}} u \leq \max_{\Gamma} u.$$

故 $\max_{\Gamma} u = \max_{\overline{Q}} u$.

□

5.7 章三第 19 题

设 $v(x, t) = z(t)u(x, t)$, 为了使用第 18 题的结论, 则 v 需要满足

$$\begin{aligned} \tilde{L}v &= v_t - a^2 v_{xx} + \tilde{c}(x, t)v \leq 0, \text{ where } \tilde{c}(x, t) \geq 0, \\ \Leftrightarrow \tilde{L}v &= z'u + zu_t - a^2 zu_{xx} + \tilde{c}(x, t)zu \\ &= z'u + zu_t - a^2 zu_{xx} + c(x, t)zu + (\tilde{c} - c)(x, t)zu \\ &= zLu + (z' + (\tilde{c} - c)z)u \\ &\leq 0. \end{aligned}$$

为保证 $Lv \leq 0$, 只需

$$\begin{cases} z \geq 0, \\ z' + (\tilde{c} - c)z = 0. \end{cases}$$

不妨取

$$\begin{cases} z = e^{c_0 t}, \\ \tilde{c} - c = -c_0. \end{cases}$$

此时

$$\tilde{L}v = e^{c_0 t} Lu \leq 0.$$

如果 $u|_{\Gamma} \leq 0$, 则 $v|_{\Gamma} = (zu)|_{\Gamma} \leq 0$, 由第 18 题结论知

$$\begin{aligned} \max_{\overline{Q}} v &= \max_{\Gamma} v \leq 0 \\ \Rightarrow \quad u|_{\overline{Q}} &\leq 0. \end{aligned}$$

□

六. PDE HOMEWORK 6

6.1 章三第 13 题

令 $v = u_t$, 则 $v \in C^{2,1}(Q) \cap C(\overline{Q})$, 且满足

$$\begin{cases} v_t - a^2 v_{xx} = f_t(x, t), & (x, t) \in Q, \\ v|_{t=0} = u_t|_{t=0} = (u_{xx} + f(x, t))|_{t=0} = \varphi''(x) + f(x, 0), & 0 \leq x \leq l, \\ v|_{x=0} = v|_{x=l} = 0, & 0 \leq t \leq T. \end{cases}$$

由解的最大模估计 p151 定理 3.2 可得

$$\begin{aligned} \max_{\overline{Q}} |u_t(x, t)| &\leq T \sup_{\overline{Q}} |f_t| + \sup_{[0, l]} |f(x, 0) + \varphi''(x)| \\ &\leq T \sup_{\overline{Q}} |f_t| + \sup_{\overline{Q}} |f(x, t)| + \sup_{[0, l]} |\varphi''(x)| \\ &\leq (1 + T) \left(\sup_{\overline{Q}} |f_t| + \sup_{\overline{Q}} |f| + \sup_{[0, l]} |\varphi''(x)| \right) \\ &:= C \left(\|f\|_{C^1(\overline{Q})} + \|\varphi''\|_{C[0, l]} \right). \end{aligned}$$

□

6.2 章三第 14 题

(1) 因为 $\varphi \in C^1[0, l]$, $\varphi(0) = u(0, 0) = 0$, 所以

$$\varphi(x) = \varphi(x) - \varphi(0) = \varphi'(\xi)x \leq \max_{[0, l]} |\varphi'(x)|x,$$

其中 $\xi \in (0, x)$, 记

$$C := \max_{[0, l]} |\varphi'(x)|, \quad (44)$$

并令 $v = Cx$, 则 v 满足

$$\begin{cases} v_t - v_{xx} = 0, & (x, t) \in Q, \\ v|_{t=0} = Cx, & 0 \leq x \leq l, \\ v|_{x=0} = 0, & 0 \leq t \leq T, \\ v|_{x=l} = Cl, & 0 \leq t \leq T. \end{cases}$$

由于

$$Lu \leq Lv, \quad (x, t) \in Q,$$

$$u|_{t=0} \leq v|_{t=0},$$

$$u|_{x=0} \leq v|_{x=0},$$

$$u|_{x=l} \leq v|_{x=l},$$

根据比较原理可得

$$u \leq v \text{ in } \overline{Q}.$$

从而

$$\frac{u(x, t) - u(0, t)}{x} \leq \frac{v(x, t) - v(0, t)}{x}.$$

令 $x \rightarrow 0^+$, 则

$$\frac{\partial u(0, t)}{\partial x} \leq C.$$

令 $v = -Cx$, 同理由比较原理可得 $u \geq v$ in $\overline{Q}$, 从而

$$\frac{u(x, t) - u(0, t)}{x} \geq \frac{v(x, t) - v(0, t)}{x}$$

令 $x \rightarrow 0^+$, 则

$$\frac{\partial u(0, t)}{\partial x} \geq -C.$$

综上可得

$$\max_{(0, T)} \left| \frac{\partial u(0, t)}{\partial x} \right| \leq C \leq \|\varphi\|_{C^1[0, l]}.$$

由 $\varphi(x) - \varphi(l) = \varphi'(\tilde{\xi})(x - l) \geq \max_{[0, l]} |\varphi'(x)|(x - l)$, 即 $\varphi(x) \geq C(x - l)$, 其中 $\tilde{\xi} \in (x, l)$ 。

同理令 $v = C(x - l)$ 或 $v = -C(x - l)$, 通过比较原理可得

$$\frac{-C(l - x)}{l - x} \leq \frac{u(x, t) - u(l, t)}{l - x} \leq \frac{C(l - x)}{l - x}.$$

令 $x \rightarrow l^-$, 则

$$\max_{(0, T)} \left| \frac{\partial u(l, t)}{\partial x} \right| \leq C \leq \|\varphi\|_{C^1[0, l]}.$$

(2) 令 $v = u_x$, 则 v 满足

$$\begin{cases} v_t - v_{xx} = 0, & (x, t) \in Q, \\ v|_{t=0} = u_x|_{t=0} = \varphi'(x), & 0 \leq x \leq l, \\ v|_{x=0} = u_x|_{x=0}, & 0 \leq t \leq T, \\ v|_{x=l} = u_x|_{x=l}, & 0 \leq t \leq T. \end{cases}$$

由解的最大模估计 p151 定理 3.2 可得

$$\begin{aligned} \max_{\overline{Q}} |u_x(x, t)| &\leq \max\left\{\sup_{[0, l]} |\varphi'(x)|, \sup_{[0, T]} |u_x(0, t)|, \sup_{[0, T]} |u_x(l, t)|\right\} \\ &\leq \max\{\|\varphi\|_{C^1[0, l]}, C\} \\ &:= \tilde{C}. \end{aligned}$$

其中 C 同 (44), 且 $\tilde{C}$ 仅依赖于 $\|\varphi\|_{C^1[0, l]}$ 。

□

6.3 章三第 16 题

记 $Q^{l_2} := \{0 < x < l_2, 0 \leq t \leq T\}$, 则 u_{l_2} 满足定解问题

$$\begin{cases} \frac{\partial u_{l_2}}{\partial t} - \frac{\partial^2 u_{l_2}}{\partial x^2} = 0, & (x, t) \in Q^{l_2}, \\ u_{l_2}|_{t=0} = 0, & 0 \leq x \leq l_2, \\ u_{l_2}|_{x=0} = g_1(t) \geq 0, & 0 \leq t \leq T, \\ u_{l_2}|_{x=l_2} = 0, & 0 \leq t \leq T. \end{cases}$$

根据弱极值原理知 $u_{l_2} \geq 0$ in Q^{l_2} , 从而 $u_{l_2}(l, t) \geq 0$ 。

记 $Q^{l_1} := \{0 < x < l_1, 0 \leq t \leq T\}$, $l_2 > l_1$, 并且令 $v = u_{l_2} - u_{l_1}$, 则 v 满足

$$\begin{cases} \frac{\partial v}{\partial t} - \frac{\partial^2 v}{\partial x^2} = 0, & (x, t) \in Q^{l_1}, \\ v|_{t=0} = 0, & 0 \leq x \leq l_2, \\ v|_{x=0} = 0, & 0 \leq t \leq T, \\ v|_{x=l_1} = (u_{l_2} - u_{l_1})|_{x=l_1} = u_{l_2}(l_1, t) \geq 0, & 0 \leq t \leq T. \end{cases}$$

同理根据弱极值原理知 $v \geq 0$ in Q^{l_1} , 即

$$u_{l_2} \geq u_{l_1}, (x, t) \in Q^{l_1}.$$

物理意义: 在长杆的一段放置热源, 另一端温度恒为 0, 热量沿着长杆传输。若杆越长, 则温度衰减越慢, 相同位置与时间下每一点温度越高。

□

6.4 章三第 17 题

(I)(i) 先证 $u(x, t) \leq u_0$ in Q , 令 $v = u_0 - u$, 则 v 满足

$$\begin{cases} \frac{\partial v}{\partial t} - \frac{\partial^2 v}{\partial x^2} = 0, & (x, t) \in Q, \\ v|_{t=0} = u_0, & 0 \leq x \leq l, \\ \left(\frac{\partial v}{\partial x} + hv\right)|_{x=0} = 0, & 0 \leq t \leq T, \\ v|_{x=l} = (u_0 - u)|_{x=l} = u_0, & 0 \leq t \leq T. \end{cases}$$

根据弱极值原理可得 $v \geq 0$ in Q , 即 $u \leq u_0$ in Q 。

(ii) 其次证 $u \geq 0$ in Q 。反设 $\exists (x_0, t_0) \in Q$, s.t. $u(x_0, t_0) < 0$, 因此 $\min_Q u(x, t) < 0$ 。由弱极值原理知

$$\min_Q u(x, t) = \min_\Gamma u(x, t),$$

其中 Γ 是 Q 的边界。不妨假设 $(x^*, t^*) \in \Gamma$, 使得 $u(x^*, t^*) = \min_\Gamma u(x, t)$, 注意到若 $x^* = l$ 则 $u(x^*, t^*) = 0$ 与假设矛盾, 因此 $x < l$, 则

$$\frac{\partial u}{\partial x}|_{(x^*, t^*)} = 0.$$

又 $u_0 > 0$, $u(x^*, t^*) < 0$, 则

$$h(u_0 - u(x^*, t^*)) > 0,$$

$$\Rightarrow \frac{u(x^*, t^*)}{\partial x} + h(u_0 - u(x^*, t^*)) > 0,$$

这与边界条件矛盾！因此 $u \geq 0$ ，综上

$$0 \leq u \leq u_0 \text{ in } Q.$$

(2) 假设对 $0 < h_1 < h_2$, u_{h_1} , u_{h_2} 满足原方程，令 $v := u_{h_2} - u_{h_1}$ ，则当 $x = 0$ 时， v 满足边界条件

$$\begin{aligned} & \left(\frac{\partial v}{\partial x} + h_2(u_0 - u_{h_2}) - h_1(u_0 - u_{h_1}) \right) \Big|_{x=0} = 0, \\ \Leftrightarrow & \left(\frac{\partial v}{\partial x} - h_1 v + (h_2 - h_1)(u_0 - u_{h_2}) \right) \Big|_{x=0} = 0, \\ \Leftrightarrow & \left(\frac{\partial v}{\partial x} + h_1 \left(\frac{h_2 - h_1}{h_1} (u_0 - u_{h_2}) - v \right) \right) \Big|_{x=0} = 0. \end{aligned}$$

代入原方程有

$$\begin{cases} \frac{\partial v}{\partial t} - \frac{\partial^2 v}{\partial x^2} = 0, & (x, t) \in Q, \\ v|_{t=0} = 0, & 0 \leq x \leq l, \\ \left(\frac{\partial v}{\partial x} + h_1 \left(\frac{h_2 - h_1}{h_1} (u_0 - u_{h_2}) - v \right) \right) \Big|_{x=0} = 0, & 0 \leq t \leq T, \\ v|_{x=l} = 0, & 0 \leq t \leq T. \end{cases}$$

由于 $h_2 - h_1 > 0$, $u_0 - u_{h_2} \geq 0$ ，由 (1) 知 $v \geq 0$ in Q ，从而 $u_h(x, t)$ 关于 h 单调递增。

物理意义：对于一根无热源的杆，其中一端的温度为 0 度，另一端与温度恒为 u_0 的介质接触，则该杆每处的温度不低于 0 度也不高于 u_0 度。若杆的热交换系数 h 越大，则杆上的每一点对应的温度越高。

□

6.5 章三第 20 题

(1) 先证 $u(x, t) \geq 0$ in Q 。令 $\tilde{u} = -u$ ，则

$$\begin{cases} \tilde{u}_t - \tilde{u}_{xx} - (\tilde{u} + a(x, t))\tilde{u} = 0, & (x, t) \in Q, \\ \tilde{u}(x, 0) = -\varphi(x) \leq 0, & 0 \leq x \leq l, \\ \tilde{u}(0, t) = \tilde{u}(l, t) = 0, & 0 \leq t \leq T. \end{cases}$$

令 $c(x, t) = \tilde{u}(x, t) + a(x, t)$ ，则 $c \in C(\overline{Q})$ ，即 $c(x, t)$ 在 $\overline{Q}$ 中有界。从而 $\exists c_0 > 0$ ，使得 $|c(x, t)| \leq c_0$ 。由章 3 第 19 题知

$$\tilde{u}|_{\Gamma} \leq 0 \Rightarrow \tilde{u}|_{\overline{Q}} \leq 0 \Rightarrow u|_{\overline{Q}} \geq 0.$$

(2) 其次证 $u \leq M \max_{[0, l]} \varphi(x)$ in Q 。令 $v(x, t) = u(x, t)z(t)$ ，且满足

$$\tilde{L}v = v_t - v_{xx} - \tilde{a}(x, t)v, \text{ where } \tilde{a}(x, t) \leq 0. \quad (45)$$

将 v 代入 (45) 得

$$\begin{aligned}\tilde{L}v &= a'u + zu_t - zu_{xx} - a(x, t)zu - (\tilde{a}(x, t) - a(x, t))zu \\ &:= zLu + z'u - (\tilde{a}(x, t) - a(x, t))zu \\ &= zLu + \left(\frac{z'}{z} - (\tilde{a}(x, t) - a(x, t))\right)v.\end{aligned}$$

为保证 $\tilde{L}v \leq 0$, 只须 $z > 0$, $\left(\frac{z'}{z} - (\tilde{a}(x, t) - a(x, t))\right) = 0$, 令 $C = \max_{\bar{Q}} |a(x, t)|$, $\Phi = \max_{[0, l]} \varphi(x)$, 并取 $\tilde{a}(x, t) = a(x, t) - C$, $z = e^{-Ct}$, 则

$$\begin{cases} \tilde{L}(v - \Phi) = e^{-Ct}Lu + (a(x, t) - C)\Phi = e^{-Ct}(-u^2(x, t)) - (C - a(x, t))\Phi \leq 0, \\ (v - \Phi)|_{\Gamma} = (e^{-Ct}u - \Phi)|_{\Gamma} \leq 0. \end{cases}$$

由章 3 第 19 题知 $v(x, t) \leq \max_{[0, l]} \varphi(x)$, 则

$$u(x, t) \leq e^{CT} \max_{[0, l]} \varphi(x).$$

□

6.6 章三第 23 题

方程两边同时乘 u 并在 $Q_\tau := Q \cap \{0 \leq t \leq \tau \leq T\}$ 上积分得

$$\frac{1}{2} \int_0^\tau \int_0^l (u^2)_t dx dt - a^2 \int_0^\tau \int_0^l uu_{xx} dx dt = \int_0^\tau \int_0^l u f dx dt. \quad (46)$$

对 (46) 进一步计算,

$$\begin{aligned}LHS &= \frac{1}{2} \int_0^l u^2(x, \tau) dx - \frac{1}{2} \int_0^l u^2(x, 0) dx - a^2 \int_0^\tau (uu_x|_0^l - \int_0^l u_x^2 dx) dt \\ &= \frac{1}{2} \int_0^l u^2(x, \tau) dx - \frac{1}{2} \int_0^l \varphi^2(x) dx + a^2 \int_0^\tau (\beta u^2(l, t) + \alpha u^2(0, t) + \int_0^l u_x^2 dx) dt \\ &\geq \frac{1}{2} \int_0^l u^2(x, \tau) dx - \frac{1}{2} \int_0^l \varphi^2(x) dx + a^2 \int_0^\tau \int_0^l u_x^2 dx dt, \\ RHS &\leq \frac{1}{2} \int_0^\tau \int_0^l u^2 dx dt + \frac{1}{2} \int_0^\tau \int_0^l f^2 dx dt.\end{aligned}$$

则

$$\frac{1}{2} \int_0^l u^2(x, \tau) dx + a^2 \int_0^\tau \int_0^l u_x^2 dx dt \leq \frac{1}{2} \int_0^\tau \int_0^l u^2 dx dt + \frac{1}{2} \int_0^\tau \int_0^l f^2 dx dt + \frac{1}{2} \int_0^l \varphi^2 dx. \quad (47)$$

令 $G(\tau) = \frac{1}{2} \int_0^\tau \int_0^l u^2 dx dt$, $F(\tau) = \frac{1}{2} \int_0^\tau \int_0^l f^2 dx dt + \frac{1}{2} \int_0^l \varphi^2 dx$, 则 (47) 可化为

$$G'(\tau) \leq G(\tau) + F(\tau).$$

又 $G(0) = 0$, $F(\tau) \geq 0$ 且单调不减, 由 Gronwall 不等式得

$$G(\tau) \leq (e^\tau - 1)F(\tau) \leq (e^T - 1)F(\tau).$$

即

$$\frac{1}{2} \int_0^\tau \int_0^l u^2 dx dt \leq \frac{e^T - 1}{2} \left(\int_0^\tau \int_0^l f^2 dx dt + \int_0^l \varphi^2 dx \right). \quad (48)$$

又由 (47) 知

$$a^2 \int_0^\tau \int_0^l u_x^2 dx dt \leq \frac{1}{2} \int_0^\tau \int_0^l u^2 dx dt + \frac{1}{2} \int_0^\tau \int_0^l f^2 dx dt + \frac{1}{2} \int_0^l \varphi^2 dx. \quad (49)$$

将 (48) 代入 (49) 得

$$\begin{aligned} \int_0^\tau \int_0^l u_x^2 dx dt &\leq \frac{1}{2a^2} \left((e^T - 1) \left(\int_0^\tau \int_0^l f^2 dx dt + \int_0^l \varphi^2 dx \right) + \int_0^\tau \int_0^l f^2 dx dt + \int_0^l \varphi^2 dx \right) \\ &= \frac{e^T}{2a^2} \left(\int_0^\tau \int_0^l f^2 dx dt + \int_0^l \varphi^2 dx \right). \end{aligned} \quad (50)$$

综上, 结合 (48) 和 (50) 得

$$\sup_{0 \leq t \leq T} \int_0^l u^2(x, t) dx + \int_0^T \int_0^l u_x^2 dx dt \leq \left(\frac{e^T}{2a^2} + e^T - 1 \right) \left(\int_0^T \int_0^l f^2(x, t) dx dt + \int_0^l \varphi^2(x) dx \right).$$

□

七. PDE HOMEWORK 7

7.1 章四第 1 题

(1) 令 $v(x) = C_0^{-1} \sup_{\Omega} |f(x)| \geq 0$, 又 $c(x) \geq C_0 > 0$, 则

$$\begin{cases} Lv = -\Delta v + c(x)v = c(x)C_0^{-1} \sup_{\Omega} |f(x)| \geq f(x) = -\Delta u + c(x)u = Lu, \\ v|_{\partial\Omega} \geq 0 = u|_{\partial\Omega}. \end{cases}$$

$$\Rightarrow \begin{cases} L(u - v) \leq 0, \\ (u - v)|_{\partial\Omega} \leq 0. \end{cases}$$

由弱极值原理知 $(u - v)$ 的非负最大值在边界上取到, 又 $(u - v)|_{\partial\Omega} \leq 0$, 则

$$\max_{\bar{\Omega}} |u(x)| \leq C_0^{-1} \sup_{\Omega} |f(x)|$$

(2) 设 $\Omega \subseteq \mathbb{R}^n$, 不妨设 Ω 包含原点, 事实上, 通过平移与重新标号即可做到. 记 $d = \text{diam}\Omega$, 模仿 p189 定理 2.5 最大模估计中的构造方法, 令

$$\begin{aligned} z(x) &= \frac{1}{2n} \sup_{\Omega} |f(x)|(d^2 - |x|^2), \\ w(x) &= z(x) \pm u(x). \end{aligned}$$

注意到 $-\Delta\left(\frac{1}{2n}(d^2 - |x|^2)\right) = 1$, $z(x) \geq 0$, 则

$$-\Delta w + c(x)w = \sup_{\Omega} |f(x)| + cz - \Delta(\pm u) + c(\pm u) \geq 0,$$

由弱极值原理知 w 在 $\bar{\Omega}$ 上的非正最小值在边界上达到, 又因为 $w|_{\partial\Omega} = z|_{\partial\Omega} \geq 0$, 故 $\forall x \in \bar{\Omega}$, $w(x) \geq 0$, 即

$$|u(x)| \leq z(x) \leq \frac{d^2}{2n} \sup_{\Omega} |f(x)|.$$

(3) 设 $\Omega = (0, \pi) \subseteq \mathbb{R}$, $u = \sin x$, $c(x) = -1$, 则 $-\Delta u - u = -u'' - u = 0$, $u(0) = u(\pi) = 0$, 若上述最大模估计成立, 则必然有 $u = 0$, 矛盾!

□

7.2 章四第 2 题

设 $F = \sup_{\Omega} |f(x)|$, $\Phi_1 = \sup_{\Gamma_1} |\varphi_1(x)|$, $\Phi_2 = \sup_{\Gamma_2} |\varphi_2(x)|$, 以及

$$\begin{aligned} z(x) &= \frac{F}{2n} \left(\frac{1 + d^2}{\alpha_0} + d^2 - |x|^2 \right), \\ w(x) &= \frac{\Phi_1}{\alpha_0} + \Phi_2 + z(x) \pm u(x). \end{aligned} \tag{51}$$

其中 $d = \text{diam}\Omega$, 不妨设 Ω 包含原点. 记 $Lw = -\Delta w + c(x)w$, 则

$$Lw = F + c(x) \left(\frac{\Phi_1}{\alpha_0} + \Phi_2 + \frac{F}{2n} \left(\frac{1 + d^2}{\alpha_0} + d^2 - |x|^2 \right) \right) \pm f(x) \geq 0.$$

设 $n = (\beta_1, \beta_2, \dots, \beta_n) = (\beta_1(x), \beta_2(x), \dots, \beta_n(x))$ 为 Ω 的单位外法向量, 则

$$\begin{aligned}\frac{\partial z}{\partial n} + \alpha(x)z &= \nabla z \cdot n + \alpha(x)z \\ &= \frac{F}{2n} \left(-2 \sum_{i=1}^n x_i \beta_i + \alpha(x) \left(\frac{1+d^2}{\alpha_0} + d^2 - |x|^2 \right) \right) \\ &\geq \frac{F}{2n} \left(-\sum_{i=1}^n (x_i^2 + \beta_i^2) + 1 + d^2 + (d^2 - |x|^2) \alpha_0 \right) \\ &\geq \frac{F}{2n} \left(-(d^2 + 1) + 1 + d^2 + (d^2 - |x|^2) \alpha_0 \right) \\ &\geq 0.\end{aligned}$$

则对 $i = 1, 2$ 有

$$\left(\frac{\partial w}{\partial n} + \alpha(x)w \right) |_{\Gamma_i} = \left(\frac{\partial z}{\partial n} + \alpha(x)w \right) |_{\Gamma_i} + \left(\alpha(x) \left(\frac{\Phi_1}{\alpha_0} + \Phi_2 \right) \right) |_{\Gamma_i} \geq 0. \quad (52)$$

若 $w(x)$ 在 $\bar{\Omega}$ 上存在负最小值, 根据弱极值原理, $w(x)$ 的负最小值 $w(x_0)$ 在 $\partial\Omega$ 上达到, 此时外法向量的方向导数

$$\begin{aligned}\frac{\partial w}{\partial n} |_{x=x_0} &= \lim_{t \rightarrow 0^+} \frac{w(x_0) - w(x_0 - tn)}{t} \leq 0, \\ \Rightarrow \left(\frac{\partial w}{\partial n} + \alpha(x)w \right) |_{x=x_0} &\leq \alpha(x_0)w(x_0) < 0,\end{aligned}$$

与 (52) 矛盾! 故 $w(x)$ 在 $\bar{\Omega}$ 上不可能有负最小值, 即 $w \geq 0$, 代入 (51) 得

$$|u(x)| \leq \frac{\Phi_1}{\alpha_0} + \Phi_2 + \frac{1}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 \right) F,$$

取 $C = \max\left\{\frac{1}{\alpha_0}, 1, \frac{1}{2n} \left(\frac{1+d^2}{\alpha_0} + d^2 \right)\right\}$ 即为所求。

□

7.3 章四第 6 题

易见 $u(x)$ 是定解问题

$$\begin{cases} -\Delta v + u^2 v = 0, & x \in \Omega, \\ \frac{\partial v}{\partial n} + \alpha(x)v = \varphi, & x \in \partial\Omega. \end{cases}$$

的解。事实上, 代入 $u = v$ 即为原方程。设 $w(x) = \frac{1}{\alpha_0} \max_{\partial\Omega} |\varphi(x)| \pm u(x)$, 则

$$\begin{aligned}-\Delta w + u^2 w &\geq 0, \\ \left(\frac{\partial w}{\partial n} + \alpha(x)w \right) |_{\partial\Omega} &\geq 0.\end{aligned} \quad (53)$$

若 $w(x)$ 在 $\bar{\Omega}$ 上存在负最小值, 根据弱极值原理, $w(x)$ 的负最小值 $w(x_0)$ 在 $\partial\Omega$ 上达到, 此时外法向量的方向导数

$$\left(\frac{\partial w}{\partial n} + \alpha(x)w \right) |_{x=x_0} \leq \alpha(x_0)w(x_0) < 0$$

与 (53) 矛盾! 故 $w(x)$ 不可能有负最小值, $w \geq 0$, 由 w 的定义得

$$\max_{\bar{\Omega}} |u(x)| \leq \frac{1}{\alpha_0} \max_{\partial\Omega} |\varphi(x)|.$$

□

7.4 章四第 7 题

根据上极限的定义, $\forall \epsilon > 0, \exists \delta_0 > 0$, 使得 $\forall x \in \partial B_{\delta_0}(P_0) \cap \Omega$

$$|u(x)| < M_0 + \epsilon.$$

任取 $x \in \Omega$, 又因为 Ω 是开集, 设 $\text{dist}(x, \partial\Omega) > d > 0$, 并取 $\delta = \min\{d, \delta_0\}$, 则 $x \in \Omega - \overline{B}_\delta(P_0)$ 。由 $-\Delta u = 0$, 在 $\Omega - \overline{B}_\delta(P_0)$ 上利用弱极值原理知 u 的最大值与最小值都在边界上达到, 从而

$$|u(x)| \leq \max\{M_0 + \epsilon, \sup_{\partial\Omega} |\varphi(x)|\}.$$

由 ϵ 与 x 的任意性得

$$\sup_{\Omega} |u(x)| \leq \max\{M_0, \sup_{\partial\Omega} |\varphi(x)|\}.$$

□

7.5 章四第 10 题

设 $|u| \leq M$ 为 $f = 0, \varphi = 0$ 时的方程的有界解, 则只需证 $u \equiv 0$ 。记 $d = \text{diam}\Omega$, 对给定任意 $\epsilon > 0$, 取充分小的 $r_0 > 0$, 使得

$$\epsilon \ln \frac{d}{r_0} > M.$$

并记 $\Omega_r = \Omega - B_r(P_0)$, 其中 $0 < r < r_0$, 考虑辅助函数

$$\begin{aligned} w(x, y) &= v(x, y) \pm u(x, y), \\ v(x, y) &= \epsilon \ln \frac{d}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}. \end{aligned}$$

注意到 v 为 Ω_r 上的调和函数。从而

$$\begin{cases} -\Delta w = -(\Delta v \pm \Delta u) = 0, & x \in \Omega, \\ w|_{\partial\Omega_r} > M - \max_{\Omega_r} |u(x)| \geq 0. \end{cases}$$

在 Ω_r 上对 w 使用弱极值原理, 则 w 的最值在边界上达到, 因此

$$w|_{\Omega_r} \geq 0,$$

由 r 的任意性得

$$w|_{\Omega} \geq 0.$$

即

$$\forall (x, y) \in \Omega, \epsilon \ln \frac{d}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} \geq |u(x, y)|,$$

再由 ϵ 的任意性得 $u \equiv 0$ 。

□

7.6 章四第 12 题

记 $F = \sup_{\Omega} |f(x)|$, $G = \sup_{\Omega} |g(x)|$, $U = \sup_{\Omega} |u(x)|$, $V = \sup_{\Omega} |v(x)|$, 考虑方程

$$\begin{cases} -\Delta u + 2u = f(x) + v, & x \in \Omega, \\ u = 0, & x \in \partial\Omega. \end{cases}$$

构造辅助函数 $h(x) = \frac{F+V}{2} \pm u(x)$ 。注意到 $-\Delta h + 2h \geq 0$, $h|_{\partial\Omega} \geq 0$, 由弱极值原理可见 $\forall x \in \Omega$, $h(x) \geq 0$, 即 $|u(x)| \leq \frac{F+V}{2}$ 。对方程

$$\begin{cases} -\Delta v + 2v = g(x) + u, & x \in \Omega, \\ v = 0, & x \in \partial\Omega. \end{cases}$$

同理有 $|v(x)| \leq \frac{G+U}{2}$, 说明

$$U \leq \frac{F+V}{2} \leq \frac{F + \frac{G+U}{2}}{2},$$

故

$$U \leq \frac{2F}{3} + \frac{G}{3} \leq \max\{F, G\},$$

同理

$$V \leq \max\{F, G\}.$$

□

7.7 章四第 14 题

只需证明在 $f \equiv 0$ 时该定解问题只有零解。对方程两端同乘 u 并在 Ω 上积分得

$$\int_{\Omega} -\Delta u \cdot u + \sum_{i=1}^n b_i(x) \frac{\partial u}{\partial x_i} u + c(x) u^2 dx = 0.$$

由 Gauss-Green 公式得

$$\begin{aligned} \int_{\Omega} -\Delta u \cdot u dx &= - \int_{\Omega} \nabla \cdot (u \nabla u) dx + \int_{\Omega} |\nabla u|^2 dx \\ &= - \int_{\partial\Omega} u \nabla u dl + \int_{\Omega} |\nabla u|^2 dx \\ &= \int_{\Omega} |\nabla u|^2 dx. \end{aligned}$$

从而

$$\int_{\Omega} |\nabla u|^2 dx + \sum_{i=1}^n b_i(x) \frac{\partial u}{\partial x_i} u + c(x) u^2 dx = 0. \quad (54)$$

利用 Young 不等式

$$\sum_{i=1}^n b_i(x) \frac{\partial u}{\partial x_i} u = \sum_{i=1}^n \frac{u b_i(x)}{\sqrt{2}} \sqrt{2} \frac{\partial u}{\partial x_i} \geq - \frac{\sum_{i=1}^n u^2 b_i^2(x)}{4} - |\nabla u|^2,$$

代入 (54) 得

$$\int_{\Omega} \left(c(x) - \frac{1}{4} \sum_{i=1}^n b_i^2(x) \right) u^2 dx \leq 0.$$

又因为 $c(x) - \frac{1}{4} \sum_{i=1}^n b_i^2(x) > 0$, 则

$$u \equiv 0.$$

□

7.8 章四第 15 题

由条件 $u \in C_0^3(\Omega)$ 可知 $u|_{\partial\Omega}, \nabla u|_{\partial\Omega}, \frac{\partial^2 u}{\partial x_i \partial x_j}|_{\partial\Omega} (i, j = 1, 2, \dots, n)$ 均为 0。反复利用 Gauss-Green 公式可得

$$\begin{aligned} \sum_{i,j=1}^n \int_{\Omega} u_{x_i x_j}^2 dx &= \sum_{i=1}^n \sum_{j=1}^n \int_{\Omega} \left(\frac{\partial^2 u}{\partial x_i \partial x_j} \right)^2 dx \\ &= \sum_{j=1}^n \int_{\Omega} \left| \nabla \frac{\partial u}{\partial x_j} \right|^2 dx \\ &= \sum_{j=1}^n \int_{\Omega} \nabla \cdot \left(\frac{\partial u}{\partial x_j} \nabla \frac{\partial u}{\partial x_j} \right) dx - \sum_{j=1}^n \int_{\Omega} \frac{\partial u}{\partial x_j} \Delta \frac{\partial u}{\partial x_j} dx \\ &= \sum_{j=1}^n \int_{\partial\Omega} \left(\frac{\partial u}{\partial x_j} \nabla \frac{\partial u}{\partial x_j} \right) \cdot \vec{n} dx - \sum_{j=1}^n \int_{\Omega} \frac{\partial u}{\partial x_j} \Delta \frac{\partial u}{\partial x_j} dx \\ &= - \sum_{j=1}^n \int_{\Omega} \frac{\partial u}{\partial x_j} \Delta \frac{\partial u}{\partial x_j} dx \\ &= - \int_{\Omega} \sum_{j=1}^n \frac{\partial u}{\partial x_j} \frac{\partial}{\partial x_j} \Delta u dx \\ &= - \int_{\Omega} \nabla u \cdot (\nabla(\Delta u)) dx \\ &= - \int_{\Omega} \nabla \cdot (\Delta u \nabla u) dx + \int_{\Omega} (\Delta u)^2 dx \\ &= - \int_{\partial\Omega} (\Delta u \nabla u) \cdot \vec{n} dx + \int_{\Omega} (\Delta u)^2 dx \\ &= \int_{\Omega} f^2 dx \\ &\leq n \int_{\Omega} f^2 dx. \end{aligned}$$

□

7.9 章四第 24 题

先证在 Ω 内每点都满足平均值性质的连续函数必在 Ω 内调和。

注意到题目中仅给出了连续性条件, 因此先证明满足平均值性质的连续函数的可微性. 考虑使用磨光函数进行卷积, 再证明其可微性, 最后证明磨光后的函数与原函数相等, 即完成可微性证明。

首先引入磨光子. 定义 η 为标准磨光子 (Standard mollifier),

$$\eta = \begin{cases} C \exp(\frac{1}{|x|^2 - 1}), & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

其中 $x \in \mathbb{R}^2$, 选定常数 $C > 0$ 使得

$$\int_{\mathbb{R}^2} \eta dx = 1,$$

则 $\eta \in C_0^\infty(\mathbb{R}^2)$. 又令

$$\eta_\epsilon(x) := \frac{1}{\epsilon^2} \eta\left(\frac{x}{\epsilon}\right),$$

则 $\int_{\mathbb{R}^2} \eta_\epsilon dx = 1$, $\eta_\epsilon \in C_0^\infty(\mathbb{R}^2)$, $\text{spt}(\eta_\epsilon) \subset \overline{B_0(\epsilon)}$. 其中 spt 为支撑集, $B_0(\epsilon)$ 为半径为 ϵ , 圆心为 0 的圆.

在 $\Omega_\epsilon := \{x \in \Omega | \text{dist}(x, \partial\Omega) > \epsilon\}$ 中定义 u^ϵ 为 η_ϵ 与 u 的卷积,

$$\begin{aligned} u^\epsilon &:= \eta_\epsilon * u \\ &= \int_{\Omega} \eta_\epsilon(x-y) u(y) dy \\ &= \int_{B_\epsilon(0)} \eta_\epsilon(y) u(x-y) dy. \end{aligned} \tag{55}$$

下边通过两个命题证明 u 的可微性.

命题 1 $u^\epsilon \in C^\infty(\Omega_\epsilon)$

固定 $x \in \Omega_\epsilon$, $i \in \{1, 2\}$, h 足够小使得 $x + he_i \in \Omega_\epsilon$, 其中 e_i 为 i 方向的单位向量, 则对于某个开的 $V \subset \subset \Omega$

$$\begin{aligned} \frac{u^\epsilon(x + he_i) - u^\epsilon(x)}{h} &= \frac{1}{\epsilon^n} \int_{\Omega} \frac{1}{h} \left(\eta\left(\frac{x + he_i - y}{\epsilon}\right) - \eta\left(\frac{x - y}{\epsilon}\right) \right) u(y) dy \\ &= \frac{1}{\epsilon^n} \int_V \frac{1}{h} \left(\eta\left(\frac{x + he_i - y}{\epsilon}\right) - \eta\left(\frac{x - y}{\epsilon}\right) \right) u(y) dy. \end{aligned} \tag{56}$$

又因为 $\eta \in C^\infty$, 则

$$\frac{1}{h} \left(\eta\left(\frac{x + he_i - y}{\epsilon}\right) - \eta\left(\frac{x - y}{\epsilon}\right) \right) \rightarrow \frac{1}{\epsilon} \eta_{x_i}\left(\frac{x - y}{\epsilon}\right)$$

在 V 上一致成立. 则由 (56) 定义的 $u_{x_i}^\epsilon(x)$ 存在并且

$$u_{x_i}^\epsilon(x) = \int_{\Omega} \eta_{x_i}(x-y) u(y) dy.$$

通过归纳法可得 $D^\alpha u^\epsilon(x)$ 存在并且

$$D^\alpha u^\epsilon(x) = \int_{\Omega} D^\alpha \eta_\epsilon(x-y) u(y) dy,$$

对 $x \in \Omega_\epsilon$, $\forall \alpha = (a_1, a_2)$ 为 2 维多重指标, $a_i \in \mathbb{N} \cup \{0\}$, $i = 1, 2$ 成立.

$$|\alpha| = a_1 + a_2, \quad D^\alpha u^\epsilon(x) := \frac{\partial^{|\alpha|} u^\epsilon(x)}{\partial x_1^{a_1} \partial x_2^{a_2}}.$$

则由 (55) 定义的 u^ϵ 的可微性得证.

命题 2 $u^\epsilon = u$

若 $x \in \Omega_\epsilon$, 则

$$\begin{aligned}
 u^\epsilon(x) &= \int_{\Omega} \eta_\epsilon(x-y)u(y)dy \\
 &= \frac{1}{\epsilon^2} \int_{B_{x_0}(\epsilon)} \eta\left(\frac{|x-y|}{\epsilon}\right)u(y)dy \\
 &= \frac{1}{\epsilon^2} \int_0^\epsilon \eta\left(\frac{r}{\epsilon}\right) \left(\int_{\partial B_{x_0}(r)} u dS \right) dr \\
 &= \frac{1}{\epsilon^2} u(x) \int_0^\epsilon \eta\left(\frac{r}{\epsilon}\right) 2\pi r dr \\
 &= u(x) \int_{B_{x_0}(\epsilon)} \eta_\epsilon dy \\
 &= u(x).
 \end{aligned}$$

则

$$\begin{aligned}
 u^\epsilon &\equiv u \text{ in } \Omega_\epsilon \\
 \Rightarrow \quad u &\in C^2(\Omega_\epsilon), \forall \epsilon > 0.
 \end{aligned}$$

综上便证明了 u 的可微性.

接下来证明 u 是调和函数. 任取 $x_0 \in \Omega$, $r > 0$, $B_{x_0}(r) \subset \Omega$, 记

$$\begin{aligned}
 f(r) &:= \oint_{\partial B_{x_0}(r)} u(x) dS \\
 &= \frac{1}{2\pi r} \int_{\partial B_{x_0}(r)} u(x) dS \\
 &= \frac{1}{2\pi} \int_{\partial B_{x_0}(1)} u(x_0 + rz) dS, \quad x = x_0 + rz \\
 &= \oint_{\partial B_{x_0}(1)} u(x_0 + rz) dS
 \end{aligned}$$

又因为 u 满足平均值性质, 则 $f(r)$ 为常数, 即

$$f'(r) \equiv 0$$

又由直接计算得

$$\begin{aligned}
 f'(r) &= \oint_{\partial B_{x_0}(1)} \nabla u(x_0 + rz) \cdot z dS \\
 &= \oint_{\partial B_{x_0}(r)} \nabla u(y) \cdot \frac{y - x_0}{r} dS \\
 &= \frac{1}{2\pi r} \int_{\partial B_{x_0}(r)} \nabla u(y) \cdot \frac{y - x_0}{r} dS \\
 &= \frac{1}{2\pi r} \int_{\partial B_{x_0}(r)} \frac{\partial u}{\partial \nu} dS \\
 &= \frac{1}{2\pi r} \int_{\partial B_{x_0}(r)} \Delta u(y) dy, \\
 &= \oint_{\partial B_{x_0}(r)} \Delta u(y) dy,
 \end{aligned}$$

其中 ν 为 $y - x_0$ 方向上的单位向量. 由 Lebesgue 微分定理, 则

$$\Delta u(x_0) = \lim_{r \rightarrow 0^+} \oint_{\partial B_{x_0}(r)} \Delta u dy = 0,$$

又因为 x_0 的任意性, 知 u 是调和函数得证.

若 u 不是常数, 且在 Ω 内达到最值, 则由强极值原理知 u 不能在 Ω 内达到非负最大值, 反之 u 为常数. 并且 $Lu \geq 0$ 说明 u 不能在 Ω 内达到非正最小值, 矛盾!

□

7.10 章四第 25 题

对 $\forall x \in \partial\Omega$, $\forall m, n > N$, $N \in \mathbb{N}$, $|u_m(x) - u_n(x)| < \epsilon$, 根据极值原理, 调和函数 $u_m - u_n$ 在 Ω 内部的值不大于其在边界的值, 因此

$$|u_m(x) - u_n(x)| < \epsilon (x \in \bar{\Omega}),$$

说明 $\{u_n\}$ 在 $\bar{\Omega}$ 上一致收敛. 设 $u_n \rightarrow u$ in $\bar{\Omega}$, 由于调和函数具有平均值性质, 故对任意 $B_P(R) \subseteq \Omega$, 有

$$u_N(P) = \frac{1}{2\pi} \int_{\partial B_P(R)} u_N(y) dl, \quad (57)$$

(57) 两边令 $N \rightarrow \infty$, 由一致收敛性得

$$\begin{aligned} u(P) &= \lim_{N \rightarrow \infty} \frac{1}{2\pi R} \int_{\partial B_P(R)} u_N(y) dl \\ &= \frac{1}{2\pi R} \int_{\partial B_P(R)} u(y) dl, \end{aligned}$$

故 u 具有平均值性质, 从而是调和函数.

□

7.11 章四第 28 题

令 $w = v - u$, 即证

$$\begin{cases} -\Delta w = 0, & (x, y) \in B(R) - O, \\ w = 0, & (x, y) \in \partial B(R). \end{cases}$$

的有界解 $w \equiv 0$. 任取 $\epsilon > 0$, 参考章四第 10 题的方法, 考虑辅助函数

$$\begin{cases} h(x, y) = f(x, y) \pm w(x, y), \\ f(x, y) := \epsilon \ln \frac{R}{\sqrt{x^2 + y^2}}. \end{cases}$$

固定 $\epsilon > 0$, 由解 v 的有界性, 取 $r_0 \in (0, R)$, 使得

$$\epsilon \ln \frac{R}{r_0} > M := \sup_{B(R)-O} |w(x, y)|.$$

记圆环 $\Omega_r = B(R) - \overline{B(r)}$, $r < r_0$, 则 f 为 Ω_r 上的调和函数, 从而在 Ω_r 上 $\Delta h = \Delta(f \pm w) = 0$ 。
在 Ω_r 上对 h 使用弱极值原理, h 在 Ω_r 内的最小值在边界上达到。注意到

$$\begin{aligned} f = w = 0, \quad (x, y) \in \partial B(R), \\ f > \epsilon \ln \frac{R}{r_0} > \sup_{B(R)-O} |w| \geq \pm w, \quad (x, y) \in \partial B(r). \end{aligned}$$

故总有 $h|_{\partial\Omega_r} \geq 0$, 从而 $h|_{\Omega_r} \geq 0$ 。由 $r < r_0$ 任意性得

$$\begin{aligned} h|_{B(R)-O} &\geq 0 \\ \Rightarrow \quad |w| &\leq \epsilon \ln \frac{R}{r}, \quad (x, y) \in B(R) - O, \end{aligned}$$

再由 ϵ 的任意性可得

$$w \equiv 0.$$

□